Forecasting for the Grain the Decision Is Made At: Price and Promotion Plans on Four Public Retail Data Sets

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Abstract. A retailer makes different decisions at different grains of the product, store and time hierarchy: store orders for one item in one store by day or week, supplier funding for an item across stores by week, category performance by store and week or across weeks, and the financial plan for the chain. Which forecasting model to use depends on the grain being solved for. This paper compares models at each of these grains on four public data sets (M5, six Dominick's categories, Breakfast at the Frat, and Corporación Favorita, the last kept out of all development), over rolling forecast origins with paired tests and model confidence sets. It addresses two of the four Ps of retail, price and promotion, through the plan inputs these data record; it does not forecast product, assortment or place, and does not solve them jointly. The models are a world model that carries each item's state forward under the price and promotion plan, the same network without that carried state, LightGBM, the Temporal Fusion Transformer, TimesFM, Chronos-Bolt, Timestral, seasonal naive, and an equal-weight average of the world model, LightGBM and TimesFM given the plan. On these data the answer changed with the grain. For an item by store and for an item across stores, the average had the lowest error on seven and eight of nine data sets, and the 90% confidence set held a median of 1 and 2 models. For a category by store, a category and the chain total, the lowest-error model differed between data sets and a median of 4 of nine models could not be told apart. For category-by-store totals over the whole window, the world model alone had the lowest cost on six of eight data sets, while it was behind LightGBM item by store on six. Every trained model used the plan, and on the weekly promotion-driven data pretrained models were less accurate without it. These are results for these data sets, with the plan inputs and constraints they carry; a forecast solved across all four Ps, or a retailer's own data, may give different results. Every model, test and change that lost is reported.

Keywords: retail forecasting; hierarchical evaluation; forecast combination; promotions; world models; foundation models; rolling-origin evaluation.

1 Introduction

A retailer uses sales forecasts for several decisions, and each decision is made at its own grain of the product, store and time hierarchy. Store orders and inventory are set for one item in one store, by day or week. Supplier funding and deal terms are agreed for an item across all stores, by week. Category performance is reviewed by category and store, week by week or over a run of weeks, and category sales and margin plans are set for the category across the chain. The financial plan is a chain total. A forecast can be accurate item by store and still miss the category total by several percent, because small errors in the same direction add up while large errors in opposite directions cancel. Knowing which grain is being solved for therefore bears on which model to use.

This paper asks that question directly: at which grain do which models forecast best, when the price and promotion plan for the weeks ahead is known? Of the four Ps of retail (product, price, promotion and place), it addresses price and promotion, through the plan inputs that public data record: shelf price, temporary price cuts, promotion flags, displays and circulars, event and payment days. It does not forecast product (assortment, new items, ranging) or place (space, distribution), and it does not solve the four together. A forecast that did, or a retailer's own data with stock, space and range recorded, could give different results from those reported here.

We compare models chosen to span what retailers use or are offered: LightGBM, the tree model behind the leading M5 entries; the Temporal Fusion Transformer; three pretrained models used without training on our data (TimesFM, Chronos-Bolt and Timestral); a world model of our own, which turns each item's recent history into a state and carries it forward one period at a time under the plan (Figure 2); the same network without the carried state; and an equal-weight average of the world model, LightGBM and TimesFM given the plan. An earlier version of this work scored one final window per data set and found that the most accurate model changed with the level scored, the month and the data set. This version tests that on rolling forecast origins taken before any window used for development, on four public data sets, one of them kept out of all development, with paired tests, model confidence sets, interval forecasts, decision costs and reconciliation, every analy-

sis written down before it ran except two checks of unexpected results, which are marked where they appear (Figure 1).

On these data sets, with the plan inputs and constraints they carry, we observed the following.

1. **The model with the lowest error depended on the grain** (§6.1). For an item by store and an item across stores, the equal-weight average had the lowest error on seven and eight of nine data sets, and few models could not be told apart from the best (median 1 and 2 of nine). For a category by store, a category and the chain total, the lowest-error model differed from data set to data set and a median of 4 models could not be told apart: at these grains the data did not settle the choice.
2. **The time grain matters too.** Summed over the whole window, the world model alone had the lowest category-by-store planning cost on six of eight data sets, although by week it was not the lowest at that grain anywhere (§6.9). For store orders at critical ratios of 0.5 and 0.8, the average or LightGBM had the lowest cost on every data set.
3. **LightGBM was the strongest single model at the fine grains, and averaging added to it.** LightGBM was more accurate than the world model item by store on six of eight data sets by the paired test. The average was more accurate than LightGBM item by store on four and never less accurate. Weights fitted on earlier windows beat equal weights in five of 54 comparisons (§6.2, §6.3).
4. **The world model added to the average on some data sets, and its carried state on almost none.** Without it the average was less accurate item by store on four data sets; on bottled juice, frozen entrees and M5 the same network without the carried state did no worse in its place (§6.3, §6.4).
5. **Every trained model used the plan, and pretrained models needed it.** Removing the plan inputs made every trained model less accurate; TimesFM with the plan was the most accurate model not trained on the data, and Timestral, released two days after this revision began, was behind it on every data set item by store and put extreme upper quantiles on a few weekly series (§6.5, §6.6).
6. **After a promotion the world model forecast more, not the dip that follows stocking up**, and the changes we tried for the dip were not kept (§6.5, §7).
7. **Every model, test and change that lost is reported**, including a seed that never reached single-fit runs and run-to-run noise as large as the differences being tested (§7).

Section 2 places the work among retail promotion models, hierarchical evaluation, forecast combination, pretrained forecasters and world models. Sections 3 to 5 give the data, the models and the experimental setup. Section 6 reports the results, Section 7 what was tried and dropped, and Sections 8 to 10 discuss what the results mean for a retailer, their limits, and conclude. The development study is in Appendix B.

2 Related work

2.1 Retail forecasting and the promotion plan

Fildes, Ma and Kolassa [5] review retail forecasting research and practice and note that most studies forecast and evaluate at one level of aggregation, usually the item, although retail decisions are spread across levels. Sales of a single item in a single store are intermittent, which led to dedicated methods for intermittent demand [6, 7] and to evaluating forecasts as count distributions [8]. The Tweedie family [9], a compound Poisson–gamma distribution with a point mass at zero, is the loss many M5 entries used, and the loss we use.

Promotions move retail sales more than anything else in the plan. PromoCast forecast promoted-week sales for a chain from the planned event and the item's past response [10]. Van Heerde, Leeflang and Wittink [11] and Macé and Neslin [12] measured the dips in sales before and after a promotion, when shoppers stock up and then buy less. Ma, Fildes and Huang [13] showed that promotional information from within and across categories improves item-level forecasts. Gupta [14] split the promotion bump into brand switching, purchase timing and quantity, and van Heerde, Gupta and Wittink [15] showed that in units only about a third of the bump is switching from other brands. Attraction models make an item's sales the category's sales times its share [16, 17]; our brand-switching step follows that structure. Dominick's Finer Foods ran pricing experiments across its stores [18], and its scanner data are behind much of this work [19].

2.2 Hierarchies, combinations and evaluation

Forecasts that must add up across a hierarchy are studied as hierarchical forecasting, from bottom-up and top-down to optimal reconciliation [21, 22, 23]. Our models forecast item by store and their forecasts are summed, as the M5 accuracy competition scored entries, with a weighted, scaled error averaged over twelve levels [1];

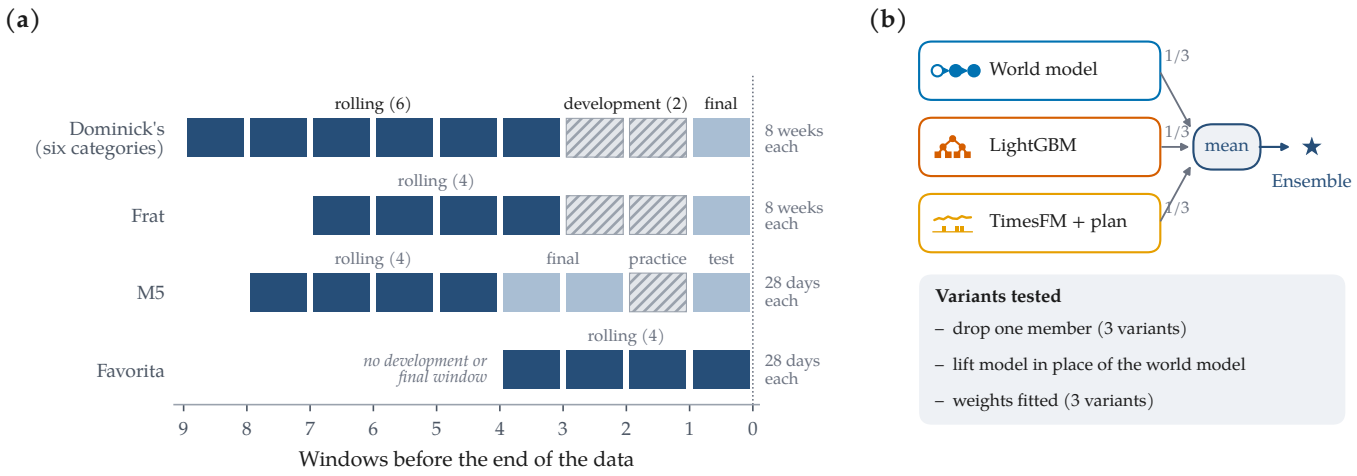


Figure 1: The main results come from rolling windows taken before any window used to choose a setting, and the average tested is fixed in advance. **(a)** Windows by data source, in periods before the end of the data: rolling windows (scored for this revision), development windows (used to choose settings) and final windows (scored once in the first version). Favorita has rolling windows only; nothing was chosen on it. **(b)** The ensemble: the equal-weight mean of the world model, LightGBM and TimesFM with the plan, and the variants of study A that test what each member adds.

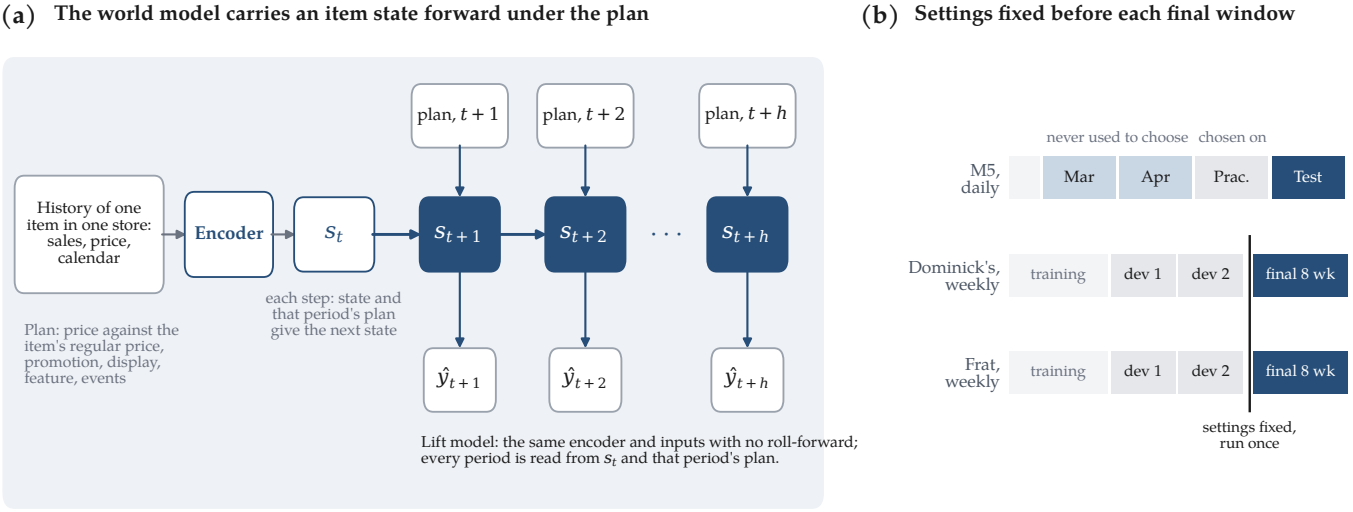


Figure 2: The world model carries each item's state forward under the plan. **(a)** An encoder turns the recent history of one item in one store into a state s_t. A transition step takes the state and one period's plan and gives the state one period later; each period's sales are read from the state as it stands after the plan so far. The lift model shares the encoder and inputs and reads every period from s_t. **(b)** On M5, settings were chosen on the practice window (25 April to 22 May 2016); on Dominick's and Frat, on two development windows before the final 8 weeks.

reconciliation is tested as a benchmark. The scaling follows Hyndman and Koehler [24], and a companion competition scored forecast distributions [2]. Tashman [25] sets out why a comparison should use several forecast origins. Hansen, Lunde and Nason's model confidence set [69] gives the set of models that cannot be told apart from the best.

Combining forecasts from different models reduces error [32, 33], and an equal-weight average is hard to beat with estimated weights, the forecast combination puzzle, which Smith and Wallis [64] explain by the error in estimating the weights; Wang and co-authors [65] review the field. We fix an equal-weight ensemble in advance and test weights fitted on earlier windows against it. Interval forecasts by calibrating each model on its own past errors follow conformal prediction [66].

2.3 Global models, trees and neural forecasters

Global models, fitted across many series at once, are often more accurate than one model per series [26]. In M5 the leading entries were built on gradient-boosted trees [27, 28]; the winner averaged direct and recursive

LightGBM models [4]. Neural global models include N-BEATS [29], DeepAR [30] and the Temporal Fusion Transformer [31], the last two taking known future inputs such as price. Graph neural networks share information between related series [34]; our item-to-item experiments use attention [59] and conditional similarity networks [35].

2.4 Pretrained foundation models

Pretrained time-series models forecast series they were not trained on. TimesFM [36] is a decoder-only transformer pretrained on a large corpus of series; Chronos [38] tokenises values and trains a language model on them, and Chronos-Bolt is its faster direct-quantile version; Lag-Llama [39], Moirai [40] and TimeGPT [41] follow similar lines. TimesFM 2.5 adds a covariate method that fits a linear model on future inputs and forecasts the remainder with the pretrained model [37]. Timestral 1.0 [68], released by Mistral on 15 October 2026, reads known future inputs inside the model itself and returns 21 quantiles. We test whether any of them, given the plan, matches models trained on it.

2.5 World models and latent prediction

A world model learns a state of its environment and how that state moves under actions, so that the effect of a planned action can be rolled forward before it is taken [42, 43, 44]. LeCun [45] proposed joint-embedding predictive architectures (JEPA), which predict the representation of the future rather than the future itself; I-JEPA [46] and V-JEPA 2 [47] apply the idea to images and video, the latter with an action-conditioned predictor used for planning. Such models guard against a collapsed representation with a slowly updated target encoder [48], a variance and covariance penalty [49], or the isotropic-Gaussian penalty of LeJEPA [50], also used in LeWorldModel [51]. TD-MPC2 [52] rolls a latent state forward under actions; DWM [53] keeps the effect of an action apart from the world's own dynamics, and AD-WM [54] trains world models to tell actions apart for counterfactual planning. Stable-worldmodel [55] evaluates world models under controlled changes, as our perturbation tests do. Closest to us, WorldTS [56] learns covariate-conditioned latent dynamics for forecasting with known future inputs. Our earlier work, M-JEPA [57], compared JEPA fine-tuning recipes with leakage-resistant splits. In retail the actions are the plan. Our world model uses a recurrent transition [58] with layer normalisation [60] on a transformer encoder [59], and is compared with the same model without the transition so that what the carried state adds is measured on its own.

Table 1: Four public data sets, all scored item by store and at every level above it. Rolling windows were added in this revision and precede every window used for development.

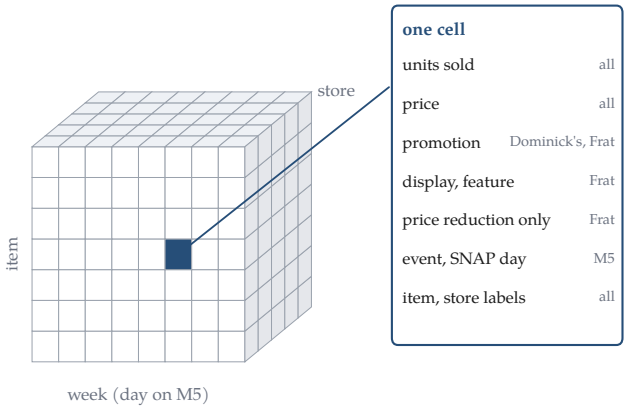
Data	Series	Plan inputs	Rolling windows
M5 (daily)	30,490	Price, event days, SNAP days	Four of 28 days, Nov 2015 to Feb 2016
Dominick's (weekly)	Six categories	Price, promotion	Six of 8 weeks per category
Frat (weekly)	3,909	Price, promotion, display, feature, price reduction only	Four of 8 weeks
Favorita (daily)	Top 10 stores	Promotion, holidays and events, paydays	Four of 28 days, the last ending 15 Aug 2017

3 Data

All four data sets are public (Table 1) and share one table shape (Figure 3a). *M5* holds daily Walmart sales of 3,049 items in 10 stores across three states from 2011 to 2016, with prices by week, event days and the days on which SNAP food assistance is paid [1]. *Dominick's Finer Foods* holds weekly scanner data by store for a Chicago chain from 1989 to 1997, with price, a promotion code and cost [19]. We use six categories chosen before any final run (cheese, snack crackers, soft drinks, bottled juice, analgesics, frozen entrees). *Breakfast at the Frat* holds weekly sales of 55 breakfast items in 77 stores from 2009 to 2012, with the shelf price, base price, and whether the item was on display, in the weekly circular (feature) or on a tag-only price reduction [20]. *Corporación Favorita* holds daily sales of an Ecuadorian grocery chain to 15 August 2017 [67]. It was added in this revision as a data set kept out of development: the series are the 10 stores with the most units sold in 2016 crossed with the items sold in all ten that year, from 1 January 2015; returns are set to zero; the plan is the promotion flag, national holidays and events, and paydays (the 15th and the last day of each month). Favorita records no prices, so its WRMSSE weights are unit weights, not revenue weights. No raw data is redistributed.

Measure. We use the weighted root mean squared scaled error (WRMSSE) of M5 [1] at each level of the hierarchy: 12 levels on M5 (Figure 3b) and six on the others. Each series' error is scaled by its own in-sample one-step error, and series are weighted by their dollar sales (units on Favorita) over the last weeks of training. We report item by store, where orders and promotions are decided, the chain total, which a financial plan is

(a) One row per item, store and period



(b) The levels a forecast is scored at, and who plans there

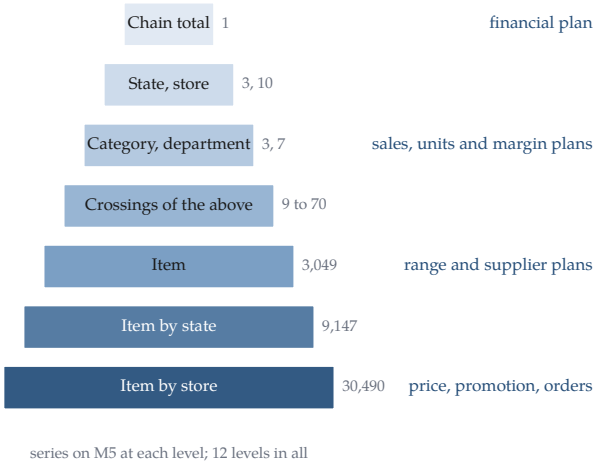


Figure 3: Every data set is one table of item, store and period, scored at every level a retailer plans at. (**a**) Each cell holds the units sold, the price and whatever plan the data set records; a lever a data set does not record is passed to the models as unknown, not as zero. (**b**) The M5 hierarchy with the number of series at each level, and the decisions usually made at each. Dominick's and Frat have six levels: total, category, store, category by store, item and item by store; Favorita has total, store, family, store by family, item and item by store.

set against, and the mean over all levels (the M5 score). Lower is better.

4 Models

Table 2 lists every model. Each trained model's technique, loss and settings are in this section and Table 3; the constraints each works within are in Appendix A.

4.1 World model

Encoder and state. For item i in one store, an encoder (a transformer over the last L periods) reads sales and price, each scaled by the item's recent level ℓ_i, together with the calendar and the recorded levers, and gives a state s_t; the item and store labels enter the readout. The item's level is its average over the history window, or, on Dominick's, over its last 8 weeks.

Transition. A gated recurrent transition f carries the state forward under the plan p_{t+h} for each future period,

$$s_{t+h}(p) = \mathrm{LN}(f(p_{t+h},\, s_{t+h-1}(p))), \quad s_t(p) = s_t, \qquad (1)$$

where LN is layer normalisation (used on M5, Dominick's and Favorita). The plan holds the period's price as a share of the item's regular price (its highest price in the last 12 periods) and the recorded levers.

Readout and loss. Expected sales are a baseline under the "nothing planned" plan p^0 (regular price, no lever)

Table 2: Every forecast compared. Plan: sees the plan for the forecast periods. Trained: fitted on the data set. Timestral is also run from history alone.

Model	Plan	Trained	Technique and loss
Ensemble	yes	yes	Equal-weight mean of world model, LightGBM, TimesFM with the plan
World model	yes	yes	Item state rolled forward under the plan; Tweedie
Lift model	yes	yes	Same network, no roll-forward; Tweedie
LightGBM	yes	yes	Direct tree model per store; Tweedie
Temporal Fusion Transformer	yes	yes	Attention over history and plan; negative binomial
Graph model (M5 final only)	yes	yes	LightGBM's inputs with message passing; Tweedie
TimesFM with the plan	yes	no	TimesFM 2.5 with its covariate method
Timestral with the plan	yes	no	Timestral-28M with known future inputs
TimesFM	no	no	TimesFM 2.5, 512 periods of units
Chronos-Bolt	no	no	Chronos-Bolt base
Seasonal naive	no	no	Sales one season earlier

times a lift,

$$\log \mu_{i,t+h} = \log \ell_i + b(s_{t+h}(p^0), p^0_{t+h}) + g(s_{t+h}(p), p_{t+h}) - g(s_{t+h}(p^0), p^0_{t+h}), \qquad (2)$$

with b and g small networks that also take the item's labels and the horizon step, and the log forecast bounded to $[-10, 10]$. The lift is exactly zero when nothing is planned, and the baseline does not move when the plan

Table 3: The world model as run in the final tests. Read from the run manifests in `results/promoted/`. Periods are days on M5 and weeks on Dominick's and Frat.

Setting	M5	Dominick's	Frat
Periods of history read	112	26	26
Periods of training data	730	156	90
State size	128	64	64
Encoder layers	3	2	2
Attention heads	4	4	4
Training steps	30000	6000	3000
Batch size	1024	512	256
Learning rate	0.001	0.001	0.001
Normalised roll-forward	yes	yes	no
Level from the last 8 weeks	no	yes	no
Brand switching	no	yes	no
Fits averaged per run	1	3	1

changes; both are checked by tests in the repository. Training minimises a Tweedie loss on sales relative to ℓ_i, with batches of item, store and date drawn at random.

Brand switching (Dominick's). In a store and week where several items are promoted, much of each item's lift is taken from the others. Following attraction models [16, 17], the world model fits on the last 8 training weeks one number per run, the share α of the summed lift that the category total keeps, and rescales its item forecasts in each store and week so that the category total is the baseline total plus α times the summed lift. α is a forecast-implied split of lift, not an identified estimate of where promoted sales come from, because promotions are not assigned at random. On Dominick's three fits with different seeds are averaged in each run.

What the state holds. A probe on the frozen world model shows that the carried state takes in the plan (Figure 4): the day's discount can be read back from the rolled state with R^2 0.58 to 0.83 across three M5 windows, against -0.01 to 0.02 from the state before the window, and still at 0.46 to 0.59 seven days later. That the plan is in the state does not show that the forecast uses it well, which §6.5 tests.

4.2 Lift model

The same model without the roll-forward: $s_{t+h}(p) = s_t$ for every h. The two share encoder, readout, size, inputs and training steps, so the difference between them is the carried state; it is the transition ablation. On Frat its readout uses dropout [61] and decoupled weight decay [62].

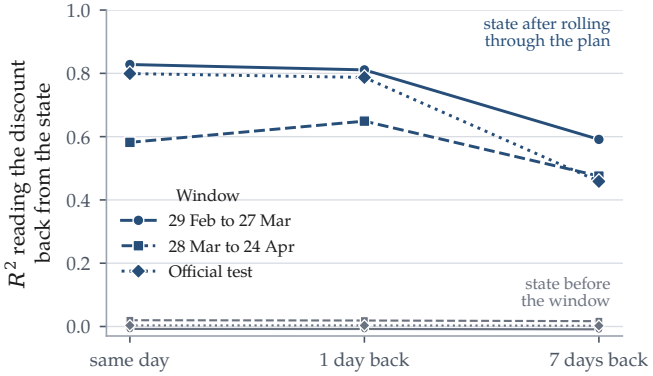


Figure 4: The rolled state holds the plan, and still holds most of it a week later. R^2 of a ridge read-out of the day's discount from the frozen world model's state, fitted on half of 6,000 item-store pairs and scored on the other half. Blue: the state after rolling through the window's plan. Grey: the state before the window. Seed 1 in three M5 windows.

4.3 Ensemble

The equal-weight mean of the forecasts of the world model, LightGBM and TimesFM with the plan for each item, store and period, fixed on 13 October 2026 before any revision result, with no weights fitted. Its intervals come from its own past errors like any other model's (§5).

4.4 Comparison models

LightGBM [27]: one direct model per store, Tweedie loss, sales at lags no shorter than the horizon, rolling averages and spread of sales, price against the item's highest recent price, calendar and plan inputs. *Temporal Fusion Transformer* [31]: one model per data set on Dominick's and Frat, fitted on the last 156 weeks with the plan as known future inputs and a negative binomial likelihood, 1,500 steps; it was not run on M5, where its training cost was beyond the compute available, nor on Favorita, whose model list was fixed without it. Chronos-Bolt and TimesFM from history alone were not run on Favorita either. *Graph model*: LightGBM's inputs with one round of GraphSAGE-style message passing [34] between the same item in other stores and items of the same department in the same store; run on the M5 final windows only. *TimesFM* [36, 37]: the 200M-parameter TimesFM 2.5 checkpoint given each series' last 512 periods; we use its mean output, because its median forecasts 27% to 31% below M5 totals (§7). *TimesFM with the plan*: the same model through its covariate method, a linear model on the plan inputs fitted over groups of 2,048 series, then TimesFM on what that model leaves. *Chronos-Bolt* [38]: the base checkpoint, zero-shot, the mean of its quantiles. *Timestral* [68]: Timestral-28M (code commit a534bdb), zero-shot, with the plan as known future inputs (price relative to the context's high-

est price, other levers as given) and from history alone; the point forecast is the mean of its 21 quantiles, each weighted by the share of probability nearest it. *Seasonal naive*: sales one season earlier.

5 Experimental setup

Windows. On Dominick's and Frat the last 8 weeks of each data set were the final window of the first version, touched once, and settings were chosen on the two 8-week windows before it. On M5 settings were chosen on the practice window (25 April to 22 May 2016), and the final windows were the official test month and, after it, the two months before the practice window. For this revision each data set gained rolling windows taken before the development windows, so none was used for any choice: six 8-week windows on each Dominick's category, four on Frat, and four 28-day windows on M5 (November 2015 to February 2016), each forecast from everything earlier (Figure 1a). Favorita has four 28-day rolling windows, the last ending on its last day, and no other.

Rules fixed in advance. The revision plan, including every window, model, test and pass rule below, was written down on 13 October 2026 before any revision run, and every change to it is a dated amendment; all but the interval fix below were made before the results they affect were read. Every model runs in every rolling window with the settings of its final configuration, unchanged. Trained models are one fit in the rolling windows: seed 1 where the configuration names a seed (M5, Frat, LightGBM everywhere), and seed 0 for the world model, lift model and TFT on Dominick's, whose configurations name none and which, before a fix on 16 October, did not pass the run's seed to the model (§7). The final windows keep their three-seed results. No new result changes a setting of a model already reported, and every model, window and test is reported.

Comparing two models. For a window and a level, the difference in WRMSSE between two models gets a 95% interval from 1,000 resamples of items: an item drawn twice counts twice in every series and aggregate it feeds, with the weights recomputed. Across windows, the mean difference gets a 95% interval from 5,000 resamples of windows. A model is said to beat another only when that interval excludes zero. For each data set and level we report the 90% model confidence set [69] by the range statistic with a bootstrap over windows.

Intervals. Each model's interval forecasts come from its own errors in the earlier rolling windows of the same data set: errors scaled by each series' scale are pooled, and their quantiles (2.5% to 97.5%) are added to the point forecast, a rolling form of conformal calibration [66]. Where a level has fewer than 50 earlier errors for each horizon, all horizons are pooled into one set of quantiles, and interval ends are rounded outwards to whole units except on Favorita; this fix was added on 16 October after the first intervals covered far less than planned at the totals, and both versions are reported. The first rolling window has no earlier window and is not scored. We report the scaled pinball loss averaged over the quantiles, weighted as in the M5 uncertainty competition [2], and coverage at 50%, 80% and 95%. Models that give quantiles of their own (Chronos-Bolt, TFT) are also scored on them, over every window.

Decisions. Three simulations turn forecasts into costs. *Store orders*: for each item-store-period, order the model's interval quantile at a critical ratio r of 0.5, 0.8 or 0.95; each unit over costs 1 and each unit short $r/(1-r)$. *Category plan*: the absolute error of each category-by-store total over the window, weighted by dollar sales. *Promotion decisions*: in the generated store worlds of the companion paper [63], where each promotion's true lift is known, promote when the predicted lift times margin exceeds a fixed cost; regret is the value lost against deciding with the true lift. The first two are reported against seasonal naive. All are simulations under stated costs, not business results.

Reconciliation. Base forecasts at every node come from AutoETS fitted on the history before each window (seasonal at the aggregate nodes, non-seasonal item by store). They are reconciled bottom-up, top-down by historical proportions, and by the MinT family [23]: ordinary least squares, weighted least squares with structural weights and with in-sample variances, and MinT with a shrunk covariance where the hierarchy has at most 6,000 nodes (Frat). Each model's own item-by-store forecasts are also combined with ETS above them by weighted least squares.

Ranking. Accuracy says how close a forecast is, not whether it puts items in the right order. For every window we report the Spearman correlation, across item-store series that sold in the window before, between the forecast change (horizon total over the previous horizon's total) and the actual change; the share of the tenth of nodes forecast to grow most that are in the actual top tenth (0.10 by chance); the Spearman correlation across promoted periods between forecast and actual units over the series' unpromoted level; and bias, forecast over actual. These were written down before they ran and have no pass rule.

Studies added after the rolling results. The rolling windows put the ensemble first, so six studies were added, each written down before it ran and reported as added after the main results: *A*, what each member adds to the ensemble and whether fitted weights beat equal ones; *B*, input perturbation on the saved models; *C* and *E*, changes aimed at the forecast after a promotion; *D*, Favorita; and *F*, Timestral, with the ranking check alongside. Three tests on the saved world models complete the set: shuffling the timing of the plan within each series, adding one promotion and price cut and following its effect through later periods, and a ridge read-out from the rolled state to sales after the window.

6 Results

All results in §6.1 to §6.9 are means over the rolling windows; §6.10 gives the final windows of the first version. Every statement is about these data sets and the plan inputs they record.

6.1 Which model at which grain

Table 4 and Figure 5 give, for each grain at which a decision is made, the model with the lowest error and how many models could not be told apart from it. Two patterns hold across the data sets.

At the fine grains the choice was settled. For an item in one store, the grain of store orders, the equal-weight average had the lowest error on seven of nine data sets and LightGBM on the other two (cheese and analgesics); the confidence set held a median of 1 model and never more than 2. For an item across all stores, the grain of supplier funding, the average was lowest on eight and LightGBM on one, with a median of 2 models in the set. Neither the world model nor any pretrained model was in the set at these two grains except TimesFM with the plan and the world model once each for an item across stores.

At the coarse grains it was not. For a category by store, a category across stores and the chain total, the lowest-error model was the average on five, five and four data sets, LightGBM on four, two and two, and the world model, the lift model or TimesFM with the plan on the rest. A median of 4 of nine models were in the confidence set at each of these grains, and up to 9: with four to six windows, these data do not separate the plan-aware models at the totals. The world model was in the set at the chain total on eight data sets and the average on all nine.

Figure 6 follows the three core models through every level of each data set's hierarchy. LightGBM stays close to the average at every level except in cheese. TimesFM with the plan moves away from the average as the grain coarsens, most on M5 and frozen entrees. The world model is closest to the average at the coarse M5 levels, where it is below it at the total, and furthest in cheese.

The time grain changes the answer again. Summed over the whole window rather than week by week, the world model had the lowest category-by-store planning cost on six of eight data sets (§6.9). The sections that follow look at each part of this picture: the fine grains and the average (§6.2, §6.3), the world model (§6.4), how the models use the plan (§6.5), the pretrained models (§6.6), the data set kept out of development (§6.7), intervals, decisions and reconciliation.

6.2 Item by store: the equal-weight average

The equal-weight ensemble has the lowest item-by-store error on seven of the nine data sets (Table 5, Figure 7a). In the other two, analgesics and cheese, LightGBM is lower: 0.647 against 0.650 in analgesics, where both are in the model confidence set, and 0.623 against 0.934 in cheese, the one data set where LightGBM is alone in the set. The ensemble is in the 90% model confidence set item by store on eight data sets and alone in it on five (snack crackers, bottled juice, Frat, M5 and Favorita). At the chain total it is in the set on all nine, and it has the lowest error on four; elsewhere the lowest belongs to LightGBM in frozen entrees; LightGBM in soft drinks; the lift model in snack crackers; TimesFM with the plan on Frat; the world model on M5.

By the paired test over windows, the ensemble is more accurate than LightGBM item by store on four of the eight data sets with paired results (bottled juice, snack crackers, Frat and M5; M5: 0.834 against 0.861) and at the total on one (analgesics). It is more accurate than the world model item by store on eight and at the total on two (cheese and Frat). It is less accurate than either at either level on none. On mean error, the Temporal Fusion Transformer, Chronos-Bolt and TimesFM from history alone are behind the ensemble item by store on every data set where they ran; they were not compared with it by a paired test.

The cheese result comes from one window, which we looked for after reading the result. In one of the six cheese windows the world model's item-by-store WRMSSE was 7.12, against a median of 0.63 over the six, and the lift model's 3.57; the ensemble, a third of whose forecast is the world model's, inherits part of that error.

6.3 What each model adds to the average

Each member earns its place on some data sets and none on all (Table 6). Without the world model the ensemble is less accurate item by store on four (bottled juice,

Table 4: On these data, which model had the lowest error depended on the grain, and the coarser the grain the more models could not be told apart. For each decision grain and data set: the model with the lowest mean WRMSSE over the rolling windows, and in brackets how many of the nine models compared are in the 90% model confidence set (models that cannot be told apart from the best). Ens.: the equal-weight ensemble of the world model, LightGBM and TimesFM with the plan; World: world model; Lift: lift model; LGB: LightGBM; TFM+p: TimesFM with the plan. On Dominick's each data set is one category, so the category and the chain total are the same series. Favorita: the category is the product family. Daily on M5 and Favorita, weekly elsewhere.

Grain	Decision made there	*Cheese*	*Snack crackers*	*Soft drinks*	*Bottled juice*	*Analgesics*	*Frozen entrees*	*Frat*	*M5*	*Favorita*
Item by store	Store orders and inventory	LGB (1)	Ens. (1)	Ens. (2)	Ens. (1)	LGB (2)	Ens. (2)	Ens. (1)	Ens. (1)	Ens. (1)
Item, all stores	Supplier funding and deals	Ens. (3)	Ens. (2)	LGB (2)	Ens. (1)	Ens. (2)	Ens. (3)	Ens. (2)	Ens. (1)	Ens. (1)
Category by store	Store-level category performance	LGB (6)	LGB (9)	Ens. (3)	Ens. (4)	Ens. (2)	LGB (4)	Ens. (2)	LGB (4)	Ens. (2)
Category, all stores	Category sales and margin plans	Ens. (5)	Lift (9)	LGB (4)	Ens. (5)	Ens. (7)	LGB (4)	Ens. (4)	World (4)	Ens. (2)
Chain total	Financial plan	Ens. (5)	Lift (9)	LGB (4)	Ens. (5)	Ens. (7)	LGB (4)	TFM+p (4)	World (4)	Ens. (3)

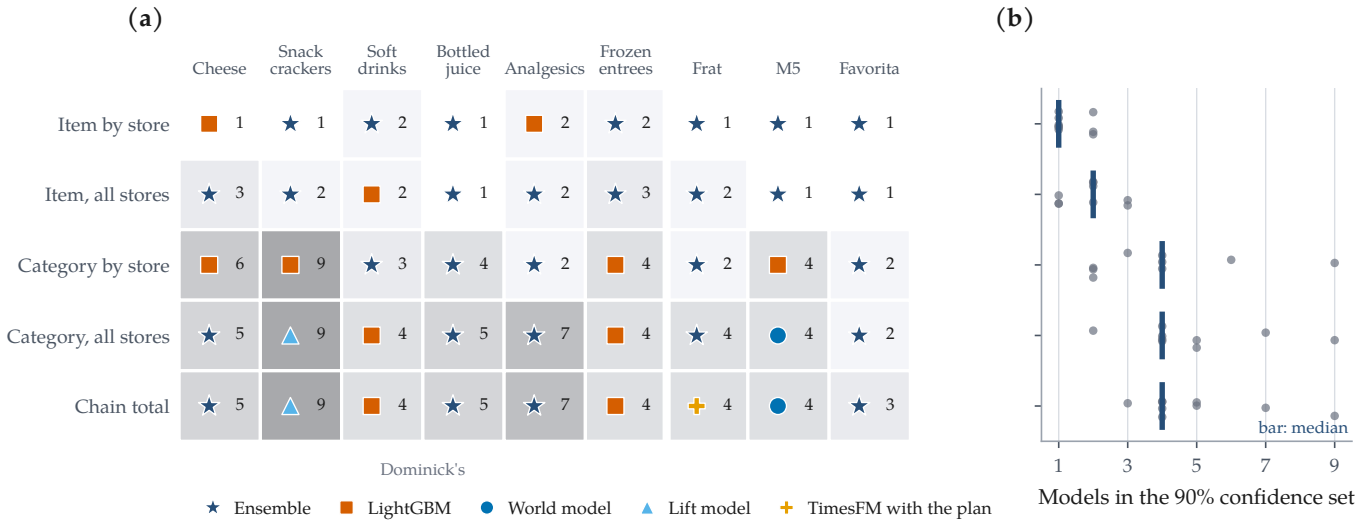


Figure 5: On these data the fine grains had a clear lowest-error model and the coarse grains did not. (**a**) For each decision grain (rows, finest at the top) and data set (columns): the model with the lowest mean WRMSSE over the rolling windows (marker) and the number of the nine models compared that are in the 90% model confidence set (number; darker cells hold more models that cannot be told apart from the best). (**b**) The size of the confidence set at each grain, one point per data set; bar: the median. On Dominick's the category and the chain total are the same series.

frozen entrees, M5 and Favorita) and at the total on three (bottled juice, frozen entrees and M5); on Frat at the total it is more accurate without it (−0.040). Without LightGBM it is less accurate item by store on eight and at the total on four. Without TimesFM with the plan it is less accurate item by store on five, at the total on one, and more accurate at the total on two (M5 and Favorita). LightGBM is the member the ensemble can least do without.

Most of the world model's contribution is any third model's. With the lift model in its place, which has the same encoder, readout, inputs and training and no carried state, the ensemble is not less accurate by the paired test than with the world model on bottled juice, frozen entrees and M5, where the world model adds item by store, and at the total on bottled juice, frozen entrees and M5. It is less accurate with the lift model item by store on three data sets (analgesics, soft drinks and Favorita) and at the total on two (analgesics and Favorita), but by the rule written down for study A the world model adds to the ensemble only where the ensemble beats the pair without it, and on soft drinks and analgesics it does not. The carried state therefore adds to an ensemble that gains from the world model only on Favorita item by

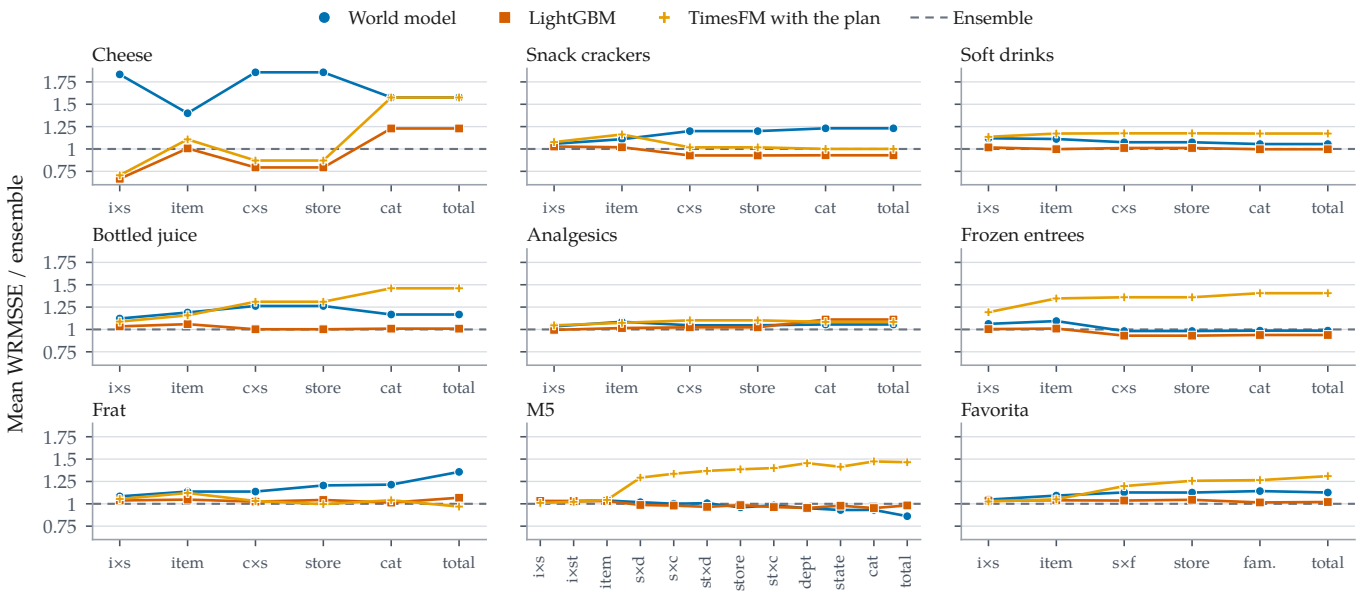


Figure 6: The three core models track the average closely at the fine grains and part ways at the coarse ones, differently on each data set. Mean WRMSSE over the rolling windows of the world model, LightGBM and TimesFM with the plan divided by the equal-weight average's (dashed line at 1), at every level of each data set's hierarchy from the finest (item by store, left) to the chain total (right). i: item; s: store; c: category; d: department; st: state; f, fam.: product family. On Dominick's and Frat the category by store equals the store and the category equals the total.

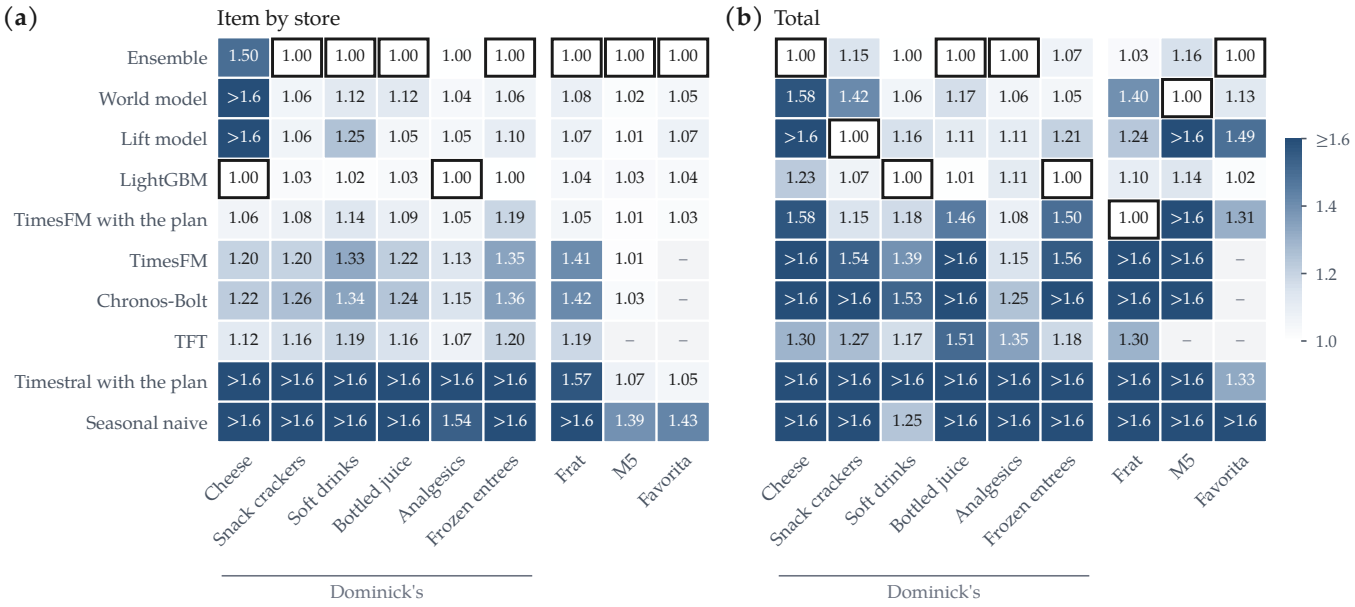


Figure 7: Item by store the average or LightGBM was best on every data set; at the total the best model changed from data set to data set. Mean WRMSSE over the rolling windows of every model divided by the best in its column (1.00: the best, outlined), **(a)** item by store and **(b)** at the chain total. Dash: not run. Six windows on each Dominick's category, four on Frat, M5 and Favorita.

store, and at the total on none.

Equal weights are hard to beat. Weights chosen on the earlier windows of each data set, for item by store, for the total or for all levels, and scored on the next window, are more accurate than equal weights in five of 54 comparisons and less accurate in 13. The fitted weights do say where each member is useful: chosen for the total, the world model's mean weight is 0.64 in analgesics and 0.62 on M5, against 0.09 and 0.32 chosen for item by store. Figure 8c shows the item-by-store error is flat across a wide range of the world model's weight on most data sets, which is why fitting the weights gains little.

Table 5: Over the rolling windows the equal-weight ensemble has the lowest item-by-store error on seven of nine data sets, is in the 90% model confidence set item by store on eight and at the total on nine. Mean WRMSSE over the rolling windows (lower is better); W: windows. Bold: the lowest in the row. °: in the 90% model confidence set of the row (Hansen, Lunde and Nason, range statistic, bootstrap over windows; Timestral, added in study F, is not in the comparison). Ensemble: the equal-weight mean of the world model, LightGBM and TimesFM with the plan, fixed before any rolling result. Trained models are one fit each (§5). The Temporal Fusion Transformer was not run on M5 or Favorita, Chronos-Bolt, TimesFM and TFT not on Favorita. Timestral+plan is Timestral-28M with the plan (study F).

Data	W	*Ensemble*	*World model*	*Lift model*	*LightGBM*	*TimesFM+plan*	*TimesFM*	*Chronos-Bolt*	*TFT*	*Timestral+plan*	*Seas. naive*
Item by store											
Cheese	6	0.934	1.713	1.119	**0.623**°	0.659	0.749	0.762	0.696	1.956	1.117
Snack crackers	6	**0.562**°	0.595	0.598	0.578	0.605	0.677	0.708	0.654	1.944	1.010
Soft drinks	6	**0.536**°	0.600	0.672	0.545°	0.609	0.711	0.720	0.640	0.872	0.960
Bottled juice	6	**0.585**°	0.656	0.616	0.605	0.636	0.712	0.728	0.676	1.459	1.079
Analgesics	6	0.650°	0.672	0.677	**0.647**°	0.679	0.734	0.745	0.693	2.111	0.997
Frozen entrees	6	**0.408**°	0.433	0.449	0.410°	0.487	0.549	0.554	0.488	1.048	0.710
Frat	4	**0.549**°	0.595	0.590	0.570	0.578	0.775	0.778	0.652	0.864	1.077
M5	4	**0.834**°	0.851	0.845	0.861	0.842	0.842	0.858	–	0.894	1.163
Favorita	4	**0.694**°	0.726	0.744	0.719	0.712	–	–	–	0.731	0.992
Total											
Cheese	6	**0.371**°	0.585°	0.617°	0.456°	0.584	0.718	0.790	0.481°	4.606	1.175
Snack crackers	6	1.181°	1.454°	**1.026**°	1.097°	1.181°	1.581°	4.410°	1.301°	16.272	1.790°
Soft drinks	6	0.834°	0.879°	0.965°	**0.831**°	0.977	1.157	1.270	0.971	4.181	1.035
Bottled juice	6	**0.525**°	0.613°	0.585°	0.530°	0.766°	0.898	0.897	0.793	6.257	1.390
Analgesics	6	**0.782**°	0.826°	0.868°	0.869°	0.848°	0.900°	0.979	1.057°	31.256	1.796
Frozen entrees	6	0.342°	0.338°	0.387°	**0.321**°	0.481	0.500	0.578	0.380	2.158	0.620
Frat	4	0.337°	0.457	0.405°	0.360°	**0.327**°	0.541	0.605	0.425	1.048	0.964
M5	4	0.706°	**0.609**°	1.163°	0.693°	1.035	1.000	1.333	–	1.025	1.087
Favorita	4	**0.303**°	0.341°	0.452	0.309°	0.397	–	–	–	0.403	0.501

6.4 The three core models, window by window

Figure 9 shows every rolling window. Item by store, the average had a lower error than LightGBM in 35 of 48 windows (median ratio 0.97), TimesFM with the plan in 12 (median 1.03) and the world model in 7 (median 1.04); the world model's worst window, in cheese, was 9.6 times LightGBM's error. At the total the world model was more accurate than LightGBM in 22 of 48 windows and TimesFM with the plan in 14, so window by window neither is clearly behind LightGBM there, while their means are (Figure 9b, c).

On its own the world model is behind LightGBM item by store and level with it at the total. By the paired test LightGBM is more accurate item by store on six of eight data sets (analgesics, bottled juice, cheese, soft drinks, snack crackers and Frat) and at the total on one (Frat); the world model is more accurate than LightGBM on none at either level. Its lowest error is at the M5 chain total, 0.609 against 0.693 for LightGBM and 0.706 for the ensemble, but the windows differ enough that the difference is not outside its interval (LightGBM minus the world model 0.085, interval −0.247 to 0.343).

Against the lift model, which differs only in the carried state, the world model is more accurate item by store on one (soft drinks), less accurate on two (bottled juice and M5), and level at the total on all eight. Taken alone, carrying the state forward neither helps nor hurts accuracy in most data sets. Its value shows in what it adds to the ensemble (§6.3) and in category plans (§6.9), where the world model has the lowest cost on six of eight data sets.

6.5 How the models use the plan

All three trained models use the plan (Table 7). With the whole plan set to "nothing planned" the total error on Dominick's rises by 116% for the world model, 97% for the lift model and 149% for LightGBM, and the forecast in planned periods changes by −59%, −58% and −58%. Price carries more of it than the promotion flag on Do-

Table 6: Removing the world model makes the ensemble less accurate item by store on four data sets and at the total on three; replacing it with the lift model, the same network without the carried state, does so on three and two. Mean WRMSSE of each variant minus the ensemble's over the rolling windows (study A, added after the rolling results); above zero the ensemble is more accurate. Bold: the ensemble is more accurate and the 95% interval from the bootstrap over windows excludes zero; underlined: the variant is, with the same test. No X: the equal-weight mean of the other two members (−World: without the world model; −LGB: without LightGBM; −TFM+p: without TimesFM with the plan). Lift: the world model replaced by the lift model. Fitted: weights chosen on the earlier windows of the data set (lowest mean WRMSSE over all levels), scored on the next, so one window fewer.

	Item by store					Total				
Data	−World	−LGB	−TFM+p	Lift	Fitted	−World	−LGB	−TFM+p	Lift	Fitted
Cheese	−0.328	**0.199**	**0.190**	−0.196	**0.015**	0.072	**0.056**	0.029	0.021	0.056
Snack crackers	0.002	**0.018**	0.004	0.002	0.004	−0.095	0.118	0.013	−0.145	−0.073
Soft drinks	0.008	**0.023**	0.009	**0.015**	−0.006	0.025	**0.045**	−0.022	−0.007	−0.008
Bottled juice	**0.012**	**0.016**	**0.015**	0.002	0.000	**0.103**	0.031	−0.042	0.051	−0.015
Analgesics	0.000	**0.014**	−0.001	**0.003**	**0.008**	0.019	−0.006	0.038	**0.025**	0.012
Frozen entrees	**0.012**	**0.019**	−0.002	0.005	−0.001	**0.025**	**0.034**	−0.026	0.013	−0.015
Frat	0.002	**0.010**	**0.012**	0.001	**0.007**	−0.040	**0.021**	**0.051**	−0.017	−0.022
M5	**0.005**	0.003	**0.006**	−0.001	**0.013**	**0.113**	0.050	−0.112	0.000	−0.037
Favorita	**0.007**	**0.006**	**0.007**	**0.010**	**0.007**	0.037	0.013	−0.015	**0.067**	0.004

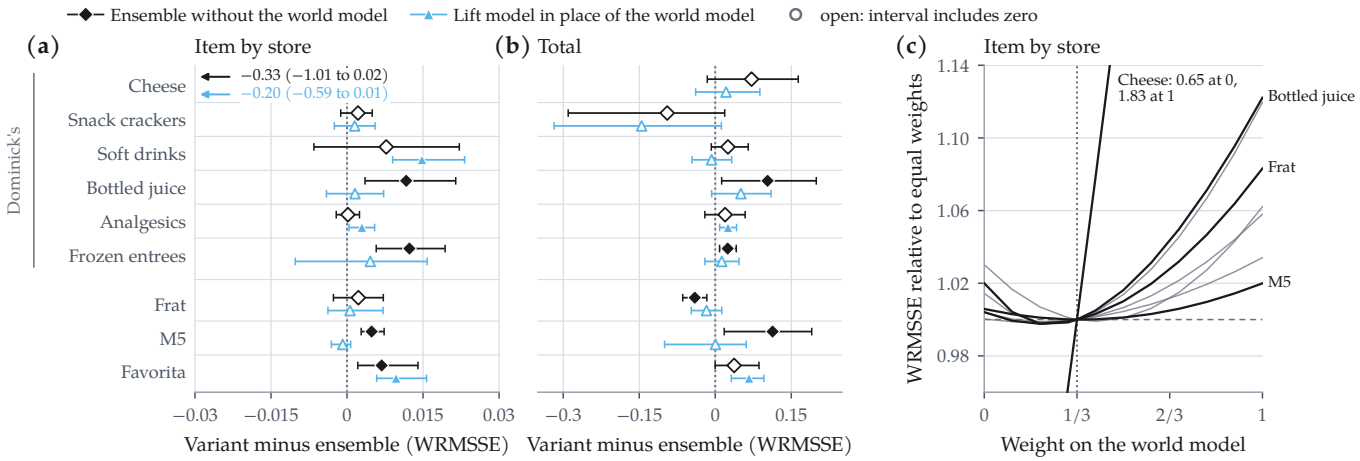


Figure 8: Taking the world model out of the ensemble costs accuracy on some data sets; replacing it with the same network without the carried state costs less often. (**a, b**) Mean WRMSSE of the ensemble without the world model, and with the lift model in its place, minus the ensemble's, with 95% intervals from the bootstrap over windows; right of zero the ensemble with the world model is more accurate; filled markers: the interval excludes zero. (**a**) Item by store. (**b**) Total. (**c**) Item by store: the ensemble's WRMSSE as the world model's weight goes from 0 to 1 (the other two sharing the rest equally), relative to equal weights (dotted line at one third), one line per data set. Study A, saved forecasts only.

minick's and Frat. On M5 prices rarely change, and the plan inputs that matter are event and SNAP days: occluding SNAP days raises the world model's total error by 58% and LightGBM's by 16%, and occluding event days by 16% against 0%. The world model reads these calendar inputs more than LightGBM does, which fits its lower error at the M5 total. On Frat the promotion flag adds little to the world model once price is given: occluding it changes the world model's error by −1% item by store and 20% at the total, against 99% for the lift model. Permuting an input within each series across the window, rather than removing it, costs more item by store and less at the total: permuting price raises the Dominick's item-by-store error by 24% for the world model and 32% for LightGBM (Table 7 shows occlusion only).

Three tests on the saved world models show what the carried state does with the plan. *Timing*: shuffling the plan within each series, with its total exposure unchanged, raises item-by-store error in cheese by 29% for the world model and 26% for the lift model, and in snack crackers by 26% and 27%: both use when the plan falls, and the carried state adds little to that. *Persistence* (Figure 10a): after a promotion and price cut added in the

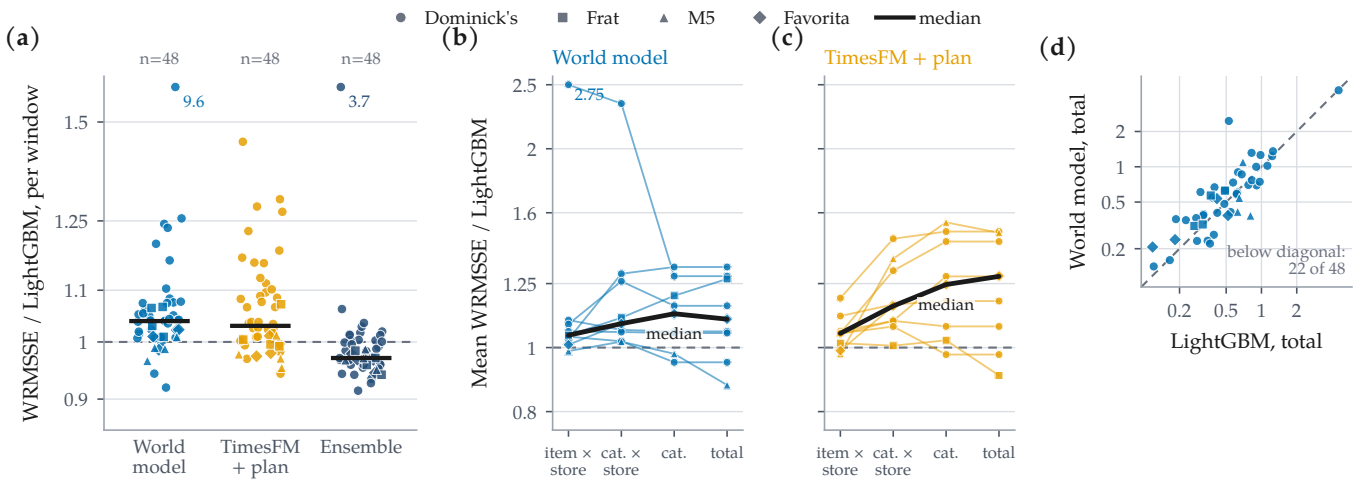


Figure 9: Window by window, the average was more accurate than LightGBM item by store in most windows, and the world model and TimesFM with the plan in few; both fell further behind LightGBM as the grain coarsened. (**a**) Item-by-store WRMSSE of the world model, TimesFM with the plan and the average, each divided by LightGBM's in the same window, one point per rolling window and data set ($n = 48$; marker: data source; bar: median; log scale; values above the axis written). (**b, c**) Mean WRMSSE over LightGBM's at four grains, one line per data set and the median (thick) for the world model and TimesFM with the plan. (**d**) Total WRMSSE of the world model against LightGBM, one point per window, log scales; below the diagonal the world model is more accurate.

Table 7: All three trained models use the plan. On the weekly data price carries most of it; on M5 the world model relies on event and SNAP days more than LightGBM does. Input occlusion on the saved models (study B): one plan input is set to "nothing planned" (regular price, lever off) in the forecast periods and the forecast recomputed, with no refitting. Change in WRMSSE item by store and at the total, in percent (above zero: the forecast gets worse without the input), and the change in forecast units, in percent, in the periods where the input was on. Mean over the rolling windows (LightGBM on M5: the last two).

	Item by store (%)			Total (%)			Units where on (%)		
Input occluded	World	Lift	LGB	World	Lift	LGB	World	Lift	LGB
Dominick's									
Whole plan	15	13	26	116	97	149	−59	−58	−58
Price	8	6	20	75	59	115	−45	−41	−49
Promotion	4	2	3	41	38	34	−34	−40	−26
Frat									
Whole plan	36	36	40	229	268	262	−49	−47	−46
Price	17	16	18	126	152	138	−38	−36	−34
Promotion	−1	6	4	20	99	34	−16	−30	−18
Feature	1	2	4	7	23	16	−28	−28	−29
Display	0	0	2	4	9	4	−18	−19	−17
Price reduction only	1	1	1	9	10	3	14	16	10
M5									
Whole plan	1	1	1	45	14	15	−8	−19	−4
Price	0	0	0	3	−26	−1	−35	−60	18
Event days	0	1	0	16	18	0	15	20	2
SNAP days	1	2	0	58	30	16	−14	−14	−6

first period, the world model's forecast stays higher in later periods, by 15% one period later in snack crackers and 8% two periods later in cheese, and above zero through the window. In categories where shoppers stock up, sales after a promotion dip; the world model carries the promotion forward as further lift. On the Frat final window, the one Frat window tested, the effect is a small dip (−2% one period later). *State*: a ridge read-out from the rolled state predicts sales in the periods after the window no better than the same read-out from the state before it (R^2 0.92 against 0.91 in cheese, 0.64 against 0.64 on M5).

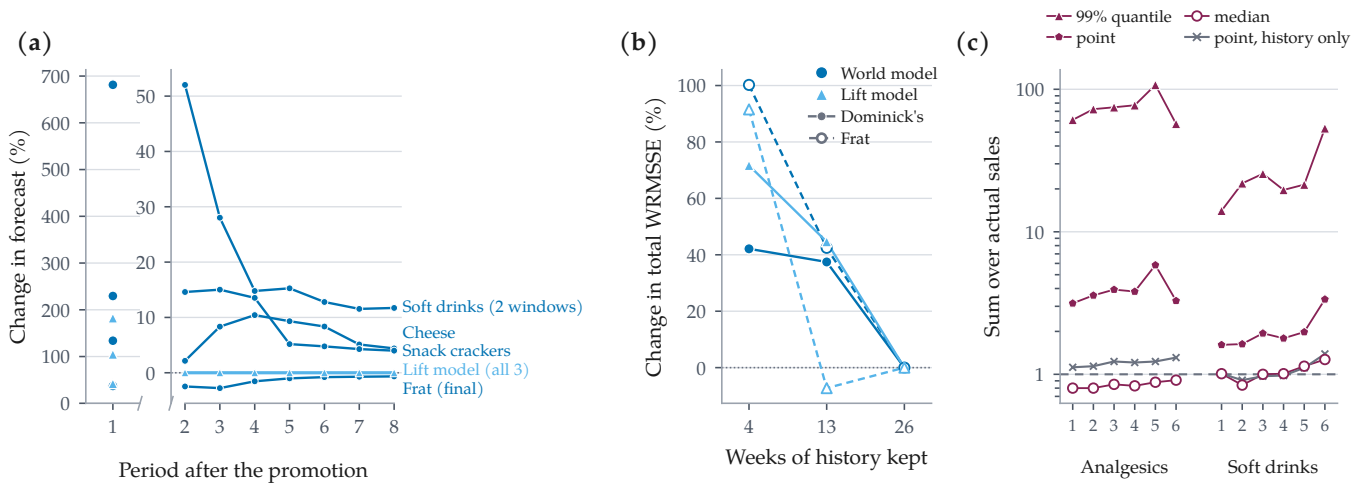


Figure 10: The world model carries a promotion forward as further lift and uses all 26 weeks of history it reads; Timestral given the plan puts extreme upper quantiles on a few series. **(a)** Persistence: the change in the forecast, in percent, at each later period after one promotion and a 20% price cut are added in the first period of the window, for the world model and the lift model (which has no carried state, so its later change is zero by construction). **(b)** History occlusion: the change in total WRMSSE, in percent, when only the last 4 or 13 of the 26 weeks of history the models read are kept (26: nothing removed). **(c)** Timestral with the plan on two Dominick's categories: the sum over all series of its median, its point forecast (the mean of its quantiles) and its 99% quantile, over actual sales, per rolling window; log scale, dashed line at 1. Study B and the mechanism tests use the saved models with no refitting; (c) from study F.

On the weekly data the world model and the lift model read the last 26 weeks of history, and removing the older part of it costs accuracy (Figure 10b): keeping only the last 13 raises the Dominick's total error by 37% for the world model and 45% for the lift model, and keeping only the last 4 by 42% and 72%. On M5, keeping only the last 28 days raises the world model's total error by 615% and the lift model's by 255%.

6.6 Pretrained models

On these data TimesFM with the plan was the most accurate model not trained on the data, and the plan matters to it: item by store it is ahead of TimesFM from history alone on every data set where both ran except M5, where they are equal to three decimals (Table 5), and most on the weekly promotion-driven data (soft drinks 0.609 against 0.711). LightGBM is more accurate than it item by store on six data sets and at the total on four, and TimesFM with the plan is more accurate than LightGBM on none. Chronos-Bolt, from history alone, is behind TimesFM from history alone on every data set item by store, and its total error in snack crackers is 4.410.

Timestral (study F, Table 8) is behind the ensemble on every data set, at both levels. Given the plan it is less accurate than TimesFM given the plan by the paired test over windows in 16 of 18 comparisons (every data set item by store and at the total), and level with it at the total on Favorita and the total on M5 (1.025 against 1.035 on M5); it is more accurate in none. From history alone it is less accurate than TimesFM from history alone in 10 of 16 comparisons and more accurate in none. This test was asked for by the plan and computed from the saved per-window scores after the first version of this section, by the bootstrap over windows only. Putting it in place of TimesFM makes the ensemble less accurate on Dominick's and Frat, level at the Favorita total and slightly less accurate item by store, and more accurate only at the M5 total (0.609 against 0.706); as written down before it ran, the paper's ensemble is unchanged and that swap is reported as a robustness result. The larger Timestral-209M, run on Frat to check the model size, is less accurate than the 28M model (1.036 against 0.864 item by store).

On Dominick's Timestral with the plan forecasts more than total sales, in analgesics several times more (Figure 10c). We checked the cause after reading the result. It is not a scale or input error: summed over all series, its median is 0.80 to 0.91 times actual sales in analgesics, and its history-only point forecast 1.12 to 1.31. Given the plan, its 99% quantile sums to 57 to 107 times actual sales, and because the point forecast is the mean of its quantiles, a few series carry the error: the top 1% of series by excess hold 75% to 88% of it. The same happens more mildly in soft drinks (99% quantile 14 to 53 times sales) and Frat (3.4 to 6.9), and not on the daily data. The forecast rule was not changed after the result. Timestral's median sums much closer to sales here, but the point forecast was fixed in advance to the mean, as for Chronos-Bolt, because a median forecasts well below

Table 8: Timestral, released two days after this revision began, is behind the ensemble on every data set, and putting it in place of TimesFM helps only at the M5 total. Mean WRMSSE over the rolling windows (Dominick's: mean of the six categories). Timestral-28M, zero-shot, the mean of its 21 quantiles as the point forecast; with the plan as known future inputs (TS+p) and from history alone (TS); TFM+p: TimesFM with the plan. Ens.: the ensemble; Ens. T: Timestral with the plan in place of TimesFM with the plan; Ens. 4: all four together. Study F, written down before it ran.

	Item by store						Total					
Data	Ens.	Ens. T	Ens. 4	TFM+p	TS+p	TS	Ens.	Ens. T	Ens. 4	TFM+p	TS+p	TS
Dominick's (six)	**0.612**	0.916	0.821	0.613	1.565	0.728	**0.672**	3.729	2.858	0.807	10.788	2.121
Frat	**0.549**	0.623	0.595	0.578	0.864	0.792	0.337	0.501	0.407	**0.327**	1.048	0.484
M5	**0.834**	0.846	0.842	0.842	0.894	0.861	0.706	**0.609**	0.699	1.035	1.025	1.014
Favorita	**0.694**	0.697	**0.694**	0.712	0.731	0.771	0.303	**0.298**	0.313	0.397	0.403	0.430

totals for items that often sell nothing; how the median would score was not tested.

6.7 A data set kept out of development

On Favorita, where nothing was tuned and every setting came from the M5 configuration, the ensemble has the lowest error at every level and is alone in the model confidence set item by store (0.694 against 0.719 for LightGBM and 0.726 for the world model); at the total LightGBM (0.309) and the world model (0.341) are in the set with it (0.303). The world model is behind LightGBM at every level, as on most of the other data sets, and still adds to the ensemble item by store: without it the ensemble's error rises by 0.007, and with the lift model in its place by 0.010. The lift model is far behind at the total (0.452). Favorita has no prices, so its weights are unit weights and its plan is promotions, holidays and paydays.

6.8 Intervals and ranking

The first conformal intervals covered far less than planned at the totals (95% intervals holding 0.27 of actual sales at the M5 total for the ensemble and 0.08 for the world model on Frat), because a level with one node has one earlier error per horizon per window, so the 2.5% and 97.5% quantiles were the smallest and largest of one to four numbers. With horizons pooled where errors are few (Table 9, Figure 11b) the ensemble's 95% interval holds 0.94 at the M5 total and 0.96 on Frat. Item by store the 95% intervals hold 0.93 to 0.98 of sales for all three models, and above 0.95 on M5 and Frat because rounding a narrow interval outwards to a whole unit widens it; the pinball loss, unchanged by rounding, is the measure to compare there, and it separates the three models by at most 0.03 on every data set. Scored on their own quantiles item by store, TFT's have a lower pinball loss than the ensemble's conformal intervals on seven of the seven data sets where it ran (0.16 against 0.18 in frozen entrees, 0.19 against 0.20 on Frat), and Chronos-Bolt's are behind them (0.42 on M5 against 0.32). Native quantiles are scored over every window and the conformal ones over all but the first, so the comparison is close but not exact.

The ensemble also ranks item-by-store changes better than any single model (Figure 11c). Across item-store series, the Spearman correlation between forecast and actual change is 0.61 on Dominick's, 0.46 on Favorita, 0.83 on Frat and 0.42 on M5, above every single model in the check on each; of the tenth of series it forecasts to grow most, 0.57, 0.51, 0.68 and 0.53 are in the actual top tenth, against 0.10 by chance. Over promoted periods LightGBM and the ensemble rank promotions best (0.67 and 0.66 on Dominick's). The ensemble's total forecast is within 6% of actual sales on every data source (0.94 on M5 is the furthest).

6.9 Decisions and reconciliation

The two decisions point to different models (Table 10, Figure 11a). For the category plan, the world model alone has the lowest cost on six of eight data sets (bottled juice, cheese, frozen entrees, soft drinks, snack crackers and M5), at 0.07 of seasonal naive's cost in frozen entrees against 0.17 for the ensemble and 0.17 for LightGBM. Its category-by-store totals over the window are close even where its item forecasts are behind LightGBM's. For store orders at a critical ratio of 0.8, LightGBM has the lowest cost on five (analgesics, cheese, frozen entrees, soft drinks and snack crackers) and the ensemble on three (bottled juice, Frat and M5); the world model on none. At 0.95 the world model is lowest on bottled juice and M5, by less than 0.02.

For promotion decisions, in the companion paper's generated worlds where the true lifts are known, deciding at the median promotion cost with the world model's predicted lift loses 1.7% to 2.1% of the value of deciding with the true lifts across three seeds, and with LightGBM's 1.8%; in the world where merchants promote into strong weeks, 2.0% to 2.5% against 1.6%. A world model that

Table 9: Rolling conformal intervals cover close to their level item by store, from 0.72 to 1.00 at the 95% level at the total, and their pinball losses item by store are within 0.03 of each other. Scaled pinball loss (SPL, lower is better) and the share of actual sales inside the 80% and 95% intervals, mean over the rolling windows after the first. Intervals from each model's own scaled errors in the earlier windows of the data set, pooled over horizons where a level has fewer than 50 earlier errors per horizon, ends rounded outwards to whole units except on Favorita (interval check of 16 October).

	Item by store									Total								
	Ensemble			World			LightGBM			Ensemble			World			LightGBM		
Data	SPL	80	95	SPL	80	95	SPL	80	95	SPL	80	95	SPL	80	95	SPL	80	95
Cheese	0.22	0.75	0.94	0.25	0.74	0.93	0.23	0.75	0.94	0.12	0.68	0.88	0.20	0.72	0.97	0.16	0.93	0.95
Snack crackers	0.23	0.74	0.94	0.24	0.73	0.94	0.23	0.73	0.93	0.53	0.60	0.72	0.80	0.57	0.60	0.42	0.75	0.88
Soft drinks	0.23	0.75	0.94	0.25	0.74	0.94	0.23	0.75	0.93	0.27	0.68	0.80	0.28	0.80	0.82	0.26	0.65	0.85
Bottled juice	0.23	0.73	0.93	0.26	0.72	0.93	0.24	0.73	0.93	0.15	0.82	0.93	0.20	0.80	0.95	0.15	0.93	0.97
Analgesics	0.25	0.91	0.98	0.25	0.91	0.98	0.24	0.91	0.98	0.25	0.85	0.88	0.29	0.85	0.95	0.27	0.88	0.95
Frozen entrees	0.18	0.69	0.93	0.18	0.69	0.93	0.17	0.70	0.93	0.12	0.90	0.97	0.12	0.88	1.00	0.10	0.82	0.90
Frat	0.20	0.87	0.97	0.22	0.87	0.97	0.21	0.87	0.97	0.11	0.75	0.96	0.15	0.83	0.88	0.14	0.62	0.83
M5	0.32	0.94	0.98	0.33	0.94	0.98	0.33	0.94	0.98	0.18	0.71	0.94	0.19	0.76	0.86	0.14	0.80	0.98
Favorita	0.24	0.79	0.95	0.25	0.79	0.95	0.24	0.79	0.95	0.07	0.83	0.98	0.10	0.70	0.93	0.07	0.86	1.00

Table 10: For category plans the world model alone has the lowest cost on six of eight data sets; for store orders the ensemble or LightGBM has the lowest cost on every one. Cost relative to seasonal naive (below 1 is better), mean over the rolling windows. Category plan: absolute error of each category-by-store total over the window, weighted by dollar sales. Store orders: order the model's 80% interval quantile for each item, store and period; each unit over costs 1 and each unit short costs 4. Simulations under stated costs, not business results. Bold: the lowest in the group.

	Category plan					Store orders (ratio 0.8)				
Data	Ens.	World	Lift	LGB	TFM+p	Ens.	World	Lift	LGB	TFM+p
Cheese	0.26	**0.12**	0.16	0.30	0.36	0.52	0.55	0.57	**0.52**	0.62
Snack crackers	0.46	**0.43**	0.50	0.46	0.48	0.72	0.77	0.77	**0.71**	0.76
Soft drinks	0.19	**0.08**	0.14	0.17	0.33	0.59	0.61	0.64	**0.57**	0.69
Bottled juice	0.23	**0.12**	0.18	0.28	0.29	**0.48**	0.49	0.51	0.49	0.54
Analgesics	0.41	0.43	0.39	**0.38**	0.42	0.68	0.70	0.71	**0.68**	0.71
Frozen entrees	0.17	**0.07**	0.19	0.17	0.27	0.51	0.52	0.60	**0.51**	0.63
Frat	**0.50**	0.57	0.56	0.51	0.52	**0.50**	0.53	0.52	0.52	0.56
M5	0.64	**0.63**	1.32	0.65	0.84	**0.82**	0.83	0.88	0.85	0.83

cannot see the promotion levers loses 77% and 98%.

Reconciliation does not replace a model trained on the plan (Table 11). The best reconciled ETS forecast on Frat is top-down, at 0.806 item by store against 0.570 for LightGBM. Combining a trained model's item forecasts with ETS above them by weighted least squares lowers the Frat total error (LightGBM 0.304 against 0.360 unreconciled; world model 0.343 against 0.457) and raises the M5 total error (world model 0.754 against 0.609). On Dominick's it raises LightGBM's total error in six of six categories (cheese 1.526 against 0.456), so it is not a fix for the weekly totals. There the best reconciled ETS forecast at the total, whichever method gives it, is behind LightGBM and the ensemble in every category. Item by store, ETS bottom-up is behind LightGBM in every category, and ahead of the world model and the ensemble only in cheese, where one window inflates their means (0.791 against 1.713 and 0.934). MinT with a shrunk covariance gave a total WRMSSE of 247.170 on Frat, a numerical failure we did not investigate further.

6.10 The final windows of the first version

The first version scored each data set once, on its final window, and its findings stand: on M5 the most accurate model depended on the level and the month (Table 12), and on Dominick's and Frat on the category (Table 13). On M5, TimesFM given only history was the most accurate single model item by store in all four months (0.885 on the test month, against 0.897 for the world model and 0.905 for LightGBM), while the plan-aware models were far more accurate at the totals. The official test month

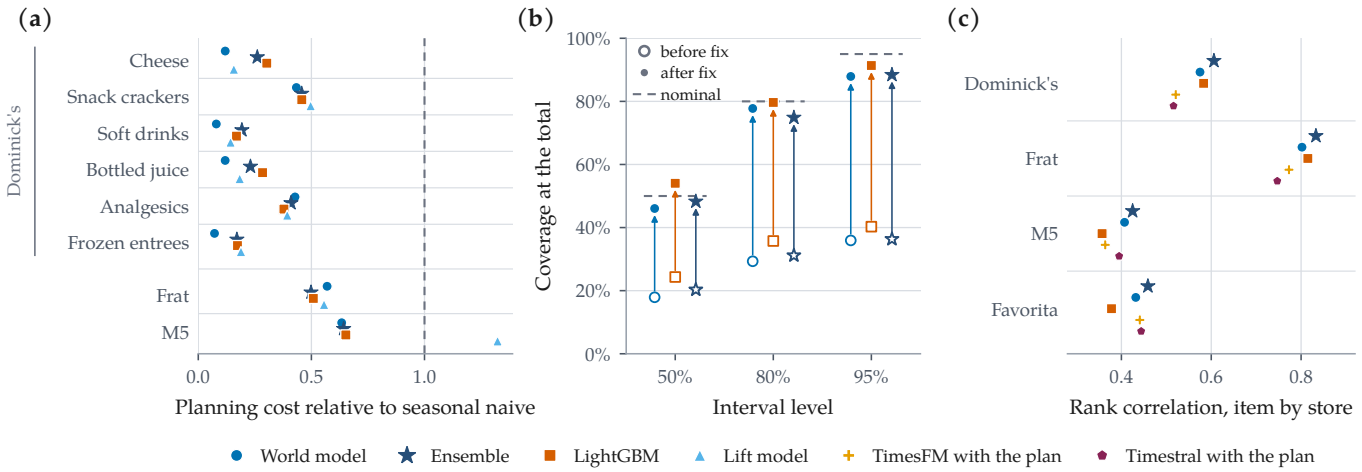


Figure 11: The world model has the lowest category-plan cost on most data sets, pooling fixed the interval coverage at the totals, and the ensemble ranks changes best. (**a**) Category-plan cost relative to seasonal naive (dashed line at 1), mean over the rolling windows. (**b**) Share of actual sales inside the 50%, 80% and 95% intervals at the chain total, mean over the eight data sets scored both before the interval fix (open) and after it (filled); dashes: the nominal level. (**c**) Spearman correlation across item-store series between forecast and actual change, mean over the rolling windows (Dominick's: over its six categories).

Table 11: Reconciled ETS forecasts are far behind the trained models; reconciling a trained model's item forecasts with ETS above them helps at the Frat total and hurts on Dominick's and M5. Mean WRMSSE over the rolling windows (Dominick's: mean of the six categories). MinT with a shrunk covariance was computed only where the hierarchy has at most 6,000 nodes; on Frat it gave a total WRMSSE of 247.170, a numerical failure not investigated further.

	Dominick's		Frat		M5	
Method	I×S	Total	I×S	Total	I×S	Total
ETS, bottom-up	0.736	1.165	0.909	0.566	0.898	1.271
ETS, top-down	0.753	0.924	0.806	0.487	0.917	0.902
ETS, OLS	0.778	1.177	0.953	0.520	0.888	0.902
ETS, WLS structural	0.748	1.125	0.912	0.538	0.885	0.908
ETS, WLS variance	0.853	1.750	0.903	0.511	0.884	0.946
ETS, MinT shrink	–	–	0.926	247.170	–	–
LightGBM + ETS, WLS variance	0.720	1.483	0.581	0.304	0.855	0.763
World model + ETS, WLS variance	0.786	1.666	0.602	0.343	0.849	0.754
LightGBM, unreconciled	0.568	0.684	0.570	0.360	0.861	0.693
World model, unreconciled	0.778	0.782	0.595	0.457	0.851	0.609
Ensemble, unreconciled	0.612	0.672	0.549	0.337	0.834	0.706

was the only one of four in which LightGBM was the most accurate over all levels (0.547 against 0.638), so a comparison on that month alone would have ranked it first. Run on the final windows (Table 14), the ensemble is lowest or within 0.02 of the lowest item by store on every data set, which agrees with the rolling windows.

How much the plan matters. On M5, where prices rarely change, giving TimesFM the plan changed its item-by-store error by at most 0.002 in any month. On Dominick's, where promoted weeks carry 18% to 61% of a category's units, TimesFM without the plan forecast 29% to 63% of promoted-week sales and with it 60% to 113% (Figure 13a). The trained plan-aware models also fell short in promoted weeks: the world model forecast 48% to 88% of promoted-week sales and LightGBM 50% to 84%.

First-time price cuts. In the April M5 window, HOUSEHOLD_1_146 had sold almost nothing for a year at 5.98, then sold 120 units in four stores over a window that held a one-week cut to 4.50 (Figure 13b). Models working from history forecast almost nothing: LightGBM 3.0, 3.0, 3.2 units across seeds, TimesFM 7.8. The world model, which reads the cut through its plan, forecast 121, 600, 243; the graph model 59, 45, 55; the lift model 79, 206,886, 316. How large the response should be is not fixed by the data: the world-model seeds differ by a factor of five,

Table 12: On M5 the best model depends on the level scored and on the month. WRMSSE on the four 28-day windows (lower is better), mean over seeds; seeds per window in the second column. All: the 12 levels together; Total: the chain total; I×S: item by store. The lowest value in each column is in bold. TimesFM with the plan has no recorded total. The lift model's April mean includes one failed seed (2.603 overall; §7).

		29 Feb to 27 Mar			28 Mar to 24 Apr			25 Apr to 22 May (practice)			23 May to 19 Jun (test)		
Model	**Seeds**	All	Total	I×S	All	Total	I×S	All	Total	I×S	All	Total	I×S
World model	3/3/3/3	**0.642**	**0.398**	0.851	**0.555**	**0.283**	0.833	0.656	0.535	0.834	0.638	0.379	0.897
Lift model	3/3/3/3	0.664	0.427	0.846	1.246	0.964	0.834	0.713	0.623	0.832	0.663	0.368	0.896
LightGBM	3/3/1/3	0.709	0.578	0.873	0.557	0.351	0.846	0.715	0.659	0.847	**0.547**	**0.247**	0.905
Graph model	3/3/3/3	0.778	0.621	0.878	0.644	0.493	0.845	**0.641**	**0.455**	0.851	0.673	0.412	0.909
TimesFM	1/1/1/1	0.854	0.830	**0.836**	0.819	0.786	**0.828**	0.896	0.930	**0.828**	0.729	0.558	0.885
TimesFM with the plan	1/1/1/1	0.858	–	0.837	0.772	–	0.829	0.929	–	0.830	0.719	–	**0.883**
Seasonal naive	1/1/1/1	0.889	0.658	1.121	0.923	0.668	1.126	0.870	0.751	1.100	0.847	0.560	1.176

Table 13: On the promotion-driven weekly data, models that see the plan lead, and which one leads changes by category. WRMSSE on the final 8 weeks (lower is better). World model, lift model and LightGBM: mean of three seeds, settings fixed on the development windows. TimesFM and TimesFM with the plan: one run each, added after the final tests. The lift model was run on Frat only. The lowest value in each row and group is in bold.

	All levels					Item by store				
Data	World	Lift	LightGBM	TimesFM	TimesFM+plan	World	Lift	LightGBM	TimesFM	TimesFM+plan
Frat	0.681	0.714	**0.549**	0.965	0.686	0.665	0.677	**0.608**	0.827	0.641
Dominick's										
Cheese	0.897	–	0.708	0.829	**0.545**	0.715	–	0.702	0.792	**0.699**
Snack crackers	0.448	–	**0.412**	0.468	0.475	0.608	–	**0.596**	0.640	0.614
Soft drinks	**0.830**	–	0.989	1.070	0.974	0.734	–	**0.652**	0.895	0.780
Bottled juice	0.987	–	0.994	**0.939**	0.990	0.816	–	**0.750**	0.782	0.754
Analgesics	**0.867**	–	0.978	1.008	0.889	**0.713**	–	0.714	0.790	0.742
Frozen entrees	0.485	–	**0.478**	0.563	0.566	0.557	–	**0.529**	0.682	0.599

Table 14: On the single final windows the ensemble is lowest or within 0.02 of the lowest item by store on every data set, and the order at the total changes from data set to data set. WRMSSE on the final window of each data set (lower is better; on M5 the official test month). One fit per trained model, so the world model and LightGBM differ slightly from the three-seed means of Tables 12 and 13. Bold: lowest in the group.

	Item by store							Total						
Data	Ens.	World	Lift	LGB	TFM+p	Chr.	TFT	Ens.	World	Lift	LGB	TFM+p	Chr.	TFT
Cheese	**0.675**	0.712	0.756	0.699	0.704	0.807	0.731	0.753	1.064	0.997	0.768	0.531	1.009	**0.522**
Snack crackers	**0.585**	0.610	0.620	0.603	0.624	0.661	0.621	0.334	0.484	0.302	0.317	0.362	0.327	**0.219**
Soft drinks	0.674	0.709	0.772	**0.655**	0.780	0.912	0.849	1.065	**0.891**	0.920	1.238	1.140	1.385	1.228
Bottled juice	0.760	0.849	0.807	0.756	**0.754**	0.785	0.775	1.111	1.019	1.190	1.171	1.172	1.046	**1.015**
Analgesics	**0.710**	0.712	0.714	0.716	0.742	0.802	0.770	1.236	1.186	1.280	1.407	1.149	1.454	**0.878**
Frozen entrees	0.543	0.574	0.563	**0.530**	0.599	0.692	0.557	0.454	0.423	**0.404**	0.438	0.537	0.536	0.414
Frat	**0.604**	0.671	0.662	0.613	0.644	0.856	0.702	0.586	0.681	0.698	**0.523**	0.677	1.222	0.818
M5	**0.881**	0.896	0.896	0.905	0.883	0.899	–	**0.240**	0.363	0.424	0.245	0.556	1.007	–

and one lift-model seed forecast over 200,000 units.

7 What was tried and dropped

We report the failures at the same length as the results. Table 15 lists what this revision tried and did not keep, and Table 16 what the first version did.

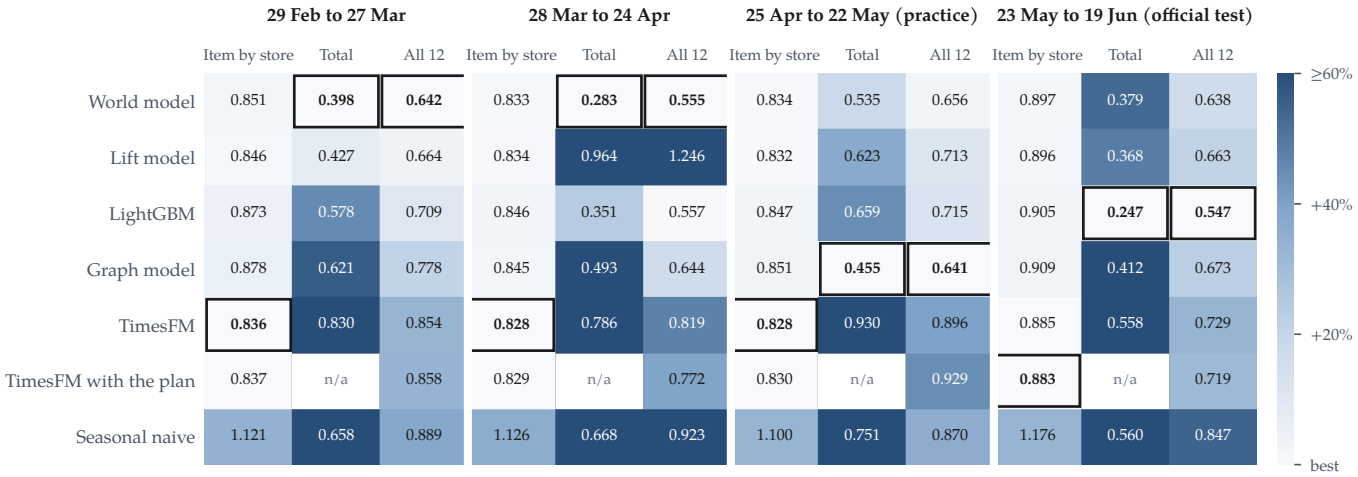


Figure 12: On the M5 final windows, TimesFM from history alone had the lowest item-by-store error of the single models in every month; at the total and over all levels the plan-aware models did, and which one changed by month. WRMSSE of every single model, four windows, three levels; cell shade: how far above the lowest value in its column (white: the lowest, outlined). Trained models: mean of three seeds. The lift model's April mean includes one failed seed (§7).

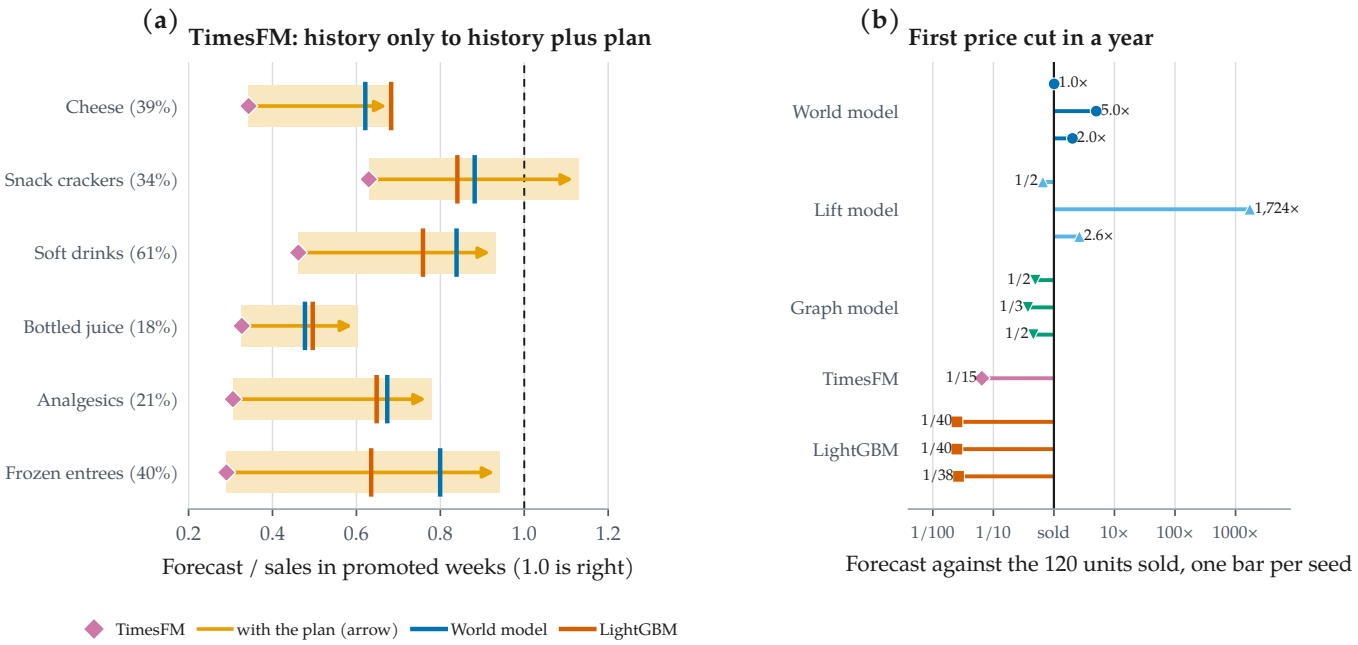


Figure 13: Given the plan, TimesFM closes most of its promoted-week shortfall; for a price cut history has not shown, plan-aware models respond and history-only models do not. **(a)** Forecast against sales in the promoted weeks of the final 8 weeks, by Dominick's category (promoted weeks' share of the category's units in brackets): TimesFM from history alone (diamond) to TimesFM with the plan (arrow head), with the world model and LightGBM (ticks). **(b)** HOUSEHOLD_1_146 in the four M5 stores where it had its first price cut in a year (5.98 to 4.50 for one week) inside the April window: each seed's forecast over the 28 days against the 120 units sold, log scale.

The forecast after a promotion. The persistence test showed the world model carrying a promotion forward as further lift where shoppers stock up. Studies C and E tried to give the dip a route. A stock term, one number per item that builds with sales above the item's level, decays by a learnt share and lowers later sales by a learnt amount that cannot be negative, made the world model worse overall in three of six categories on the development windows. On the lift model it passed the development rule, but on the rolling windows it did not fix the after-promotion forecast (soft drinks 1.27 with it, 1.28 for the plain lift model, 1.19 for the world model). A loss weight on the periods after a promotion lowered the error there in all six categories and made the overall score worse in four. None became a default. The reason given for study C also needed a correction: the 1.32 and 1.33 after-promotion ratios that prompted it came from an earlier world-model version; the final world model was

Table 15: What the revision tried and did not keep. Each was written down with its pass rule before it ran. dev.: the development windows of the six Dominick's categories. After promotion: forecast over actual in the two periods after a promotion.

Change or model	Where tested	Result
Stock term on the world model (study C)	Dominick's dev., seed 1	Overall worse than the world model in three of six categories
Plan of the two periods before, on the lift model (study C)	Dominick's dev., seed 1	Overall worse than the world model in three of six
Stock term on the lift model (study C)	Dominick's dev., then rolling	Passed on development (after promotion 0.97 soft drinks, 0.93 snack crackers); on the rolling windows 1.27 and 1.16, against 1.19 and 1.17 for the world model
Loss weight $1 + w$, $w = 2$, after a promotion (study E)	Dominick's dev., seed 1	Error after promotion lower in all six (soft drinks 0.174 to 0.150); overall worse in four (snack crackers 0.749 to 0.857)
Loss weight $1 + w$, $w = 5$ (study E)	Dominick's dev., seed 1	Overall worse in four; snack crackers after promotion 1.11
Fitted ensemble weights (study A)	All nine, rolling	Better than equal weights in five of 54 comparisons, worse in 13
Timestral in place of TimesFM (study F)	All four sources, rolling	Worse on Dominick's and Frat; better at the M5 total only
Timestral-209M (study F)	Frat, rolling	Less accurate than 28M; too slow for the other data sets
MinT, shrunk covariance	Frat, rolling	Total WRMSSE 247.170; not investigated

Table 16: Every change and comparison model that did not become a default. Overall WRMSSE unless stated; "against" gives the default it was compared with on the same window. dev.: development windows before the final 8 weeks. Practice: the M5 window 25 April to 22 May 2016. Each row is a section of `results/README.md` with its run folders.

Change or model	Where tested	Result
Twice the training steps	Dominick's (6) and Frat, dev., 3 seeds	Better in 2, worse in 3, level or mixed in 2
Weekday as seven inputs	M5 practice, seed 1	0.653 against 0.646
Event kinds (with weekday)	M5 practice, seed 1	0.646 against 0.646
60,000 training steps	M5 practice, seed 1	0.642 against 0.646; gain below seed range 0.029
Item level against past-year level	M5 practice, seed 1	0.680 against 0.646
Totals term in the loss	Dominick's analgesics, dev.	0.819 against 0.683; total 8% under
Negative binomial likelihood	Frat, analgesics, dev.	Level
Plan split, latent check, near-week weighting	Frat, dev., 5 seeds	0.417 to 0.427 against 0.408
Own forecast fed into the roll-forward	Dominick's (6), dev.	Worse overall in 4 of 6
Item role labels	Dominick's (6), dev.	No effect on LightGBM; snack crackers 2.422 for the world model
Averaging three fits	M5 practice, seed 1	0.665 against 0.656 (kept on Dominick's)
Graph model over product attributes	M5, four windows, 3 seeds	Level with LightGBM item by store
TimesFM, median output	M5, four windows	Overall 1.71 to 1.88; totals 27% to 31% under
Lift model (no roll-forward)	M5, four windows, 3 seeds	One seed failed in April (2.603)

at 1.00 and 1.04 on those windows.

Seeds and run-to-run noise. Stage 2 of study C repeated each configuration with seeds 2 and 3 and got identical scores in all twelve runs. Single-fit configurations whose settings named no seed trained with seed 0 for every run seed; the seed now reaches the model, and the results already recorded are those of seed 0 (each manifest records its commit). The repeat also showed that two runs of the same configuration on different Colab machines differ by as much as the variants being tested: the world model's overall score in snack crackers was 0.911 on one machine and 0.749 on another. Study E therefore reran the world model in the same job as the variants it was compared with. The rolling windows, being one fit per model, carry this noise; the paired tests over windows absorb part of it, not all.

A lift-model seed that broke on five series. In the April M5 final window, lift-model seed 2 scored 2.603 over all levels against 0.566 and 0.570 for seeds 1 and 3, and forecast 1,413,724 units against 1,183,626 sold. Five item-store series hold all of the excess, each with its first price cut in a year inside the window; the seed read the cuts as lifts up to the model's bound. Item by store, weighted by dollar sales, the failure is invisible (0.834); at the totals it dominates. The world model did not fail on these series in any seed.

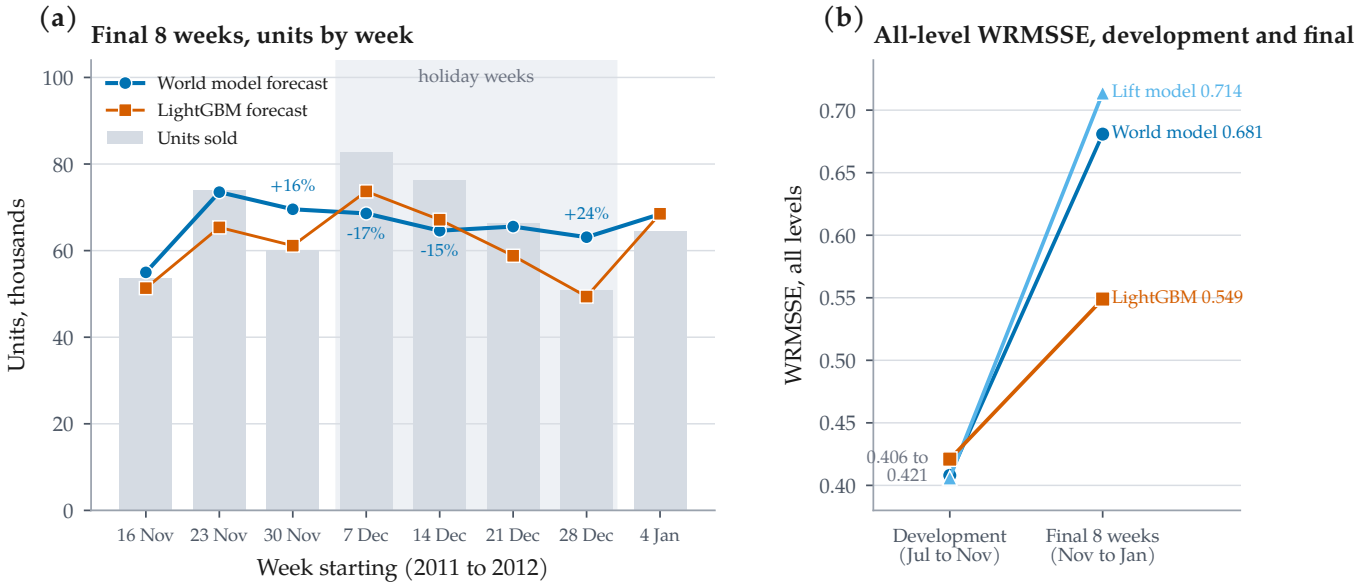


Figure 14: On the Frat final window the world model got the 8-week total right and missed the holiday shape, which the development windows did not contain. (**a**) Units sold each week of the final window over all item-stores (bars), and the weekly forecasts of the world model and LightGBM, three seeds averaged; labels: the world model's largest weekly misses. (**b**) All-level WRMSSE on the development windows and on the final 8 weeks.

Frat holiday weeks. On the Frat development windows the world model, lift model and LightGBM were level (0.406 to 0.421). On the final 8 weeks the world model fell to 0.681 against 0.549 for LightGBM. The final weeks are the holiday period, which the development windows did not contain (Figure 14); the world model's 8-week total is right (forecast against sales 1.000) but it is 15% to 17% under in the two December peak weeks. LightGBM, with week-of-year and month inputs and the previous holiday season in its training data, follows that shape more closely.

TimesFM's default output. TimesFM's default point forecast is the median. For an item that sells nothing on most days the median is near zero while the mean is not, so the median forecast is 27% to 31% below M5 totals in every month. The mean output, used throughout, does not have the problem.

8 Discussion

The grain a decision is made at bore on which model forecast it best on these data (Table 17). The rows below are observations on four public data sets with price and promotion plans, not a ranking that holds for every retailer: a retailer's own data, with stock, space and range recorded, or a forecast solved across all four Ps, could order the models differently, and the comparison should be run there.

Why an equal-weight average. On these data the three members failed in different places. LightGBM is the most accurate single model item by store on most data sets and the member the ensemble can least do without. TimesFM with the plan brings a pretrained view of each series' history. The world model has the lowest category-plan cost on six of eight data sets, and the ensemble without it is less accurate item by store on four. An average of three models whose errors differ has a lower mean error item by store than any of them on seven of nine data sets, and by the paired test beats LightGBM on four of eight; equal weights are as good as weights fitted on a few earlier windows, as the forecast combination literature leads one to expect [64, 65].

What the carried state adds. Against the lift model, which differs only in the carried state, the world model is level on its own in most data sets, adds to the ensemble and beats its lift-model replacement there only on Favorita item by store, and has a lower category-plan cost than the lift model on six of eight. The carried state holds the plan (Figure 4), but the rolled state does not predict sales beyond the window better than the state before it, and it carries a promotion forward as more lift rather than a dip. On these data the case for the carried state rests on category totals over the window, not on item accuracy, and its contribution to the average was mostly that of any third model.

Two cautions. A plan-driven forecast responds to plans unlike history with a size the data do not fix: one lift-model seed and Timestral's upper quantiles both pro-

Table 17: Each decision has its own grain, and on these data the model that forecast it best differed by grain. Observations on the four public data sets of this paper.

Decision	Grain	What was observed on these data
Store orders and inventory	Item by store, day or week	The average lowest error on seven of nine, LightGBM on two; one or two models in the confidence set; the average or LightGBM lowest order cost on every data set at critical ratios 0.5 and 0.8
Supplier funding, promotion and price decisions	Item across stores, by week, under a planned change	The average lowest error on eight of nine; a model that reads the plan is needed, since history alone forecast 29% to 63% of promoted-week sales; after a promotion the world model forecast more rather than the dip
Category performance by store	Category by store, by week	No settled choice: the average lowest on five, LightGBM on four; a median of 4 models in the confidence set
Category sales and margin plans	Category by store over the window; category across stores	Over the window the world model alone had the lowest planning cost on six of eight; by week no settled choice (median 4 models in the set)
Financial plan	Chain total	No settled choice; the average in the confidence set on all nine
Choosing a model	Any	Score several windows with paired tests; one test month reversed the ranking on M5, and runs on different machines differed by as much as the changes tested
Adopting a new pretrained model	Any	Check its totals and its upper quantiles before use; Timestral's tails made its mean forecast 3 to 6 times sales in analgesics

duced forecasts many times any sale. And a forecast that is accurate item by store can be several percent off the category total, which is the number a merchant plans against.

9 Limitations and threats to validity

Scope: two of the four Ps, on public data. The paper addresses price and promotion. It does not forecast product (assortment, new items, ranging) or place (space, distribution), and it does not solve the four Ps together; a unified forecast across them does not exist in this work, and one could change which model works best at each grain. The data are four public data sets with their own plan inputs and constraints. A single retailer's data, with its own levers, stock and space records, could give a different ordering of models, and the grain comparison should be repeated on it before a choice is made.

Windows and seeds. Trained models are one fit each in the rolling windows, and runs of one configuration on different machines differed by as much as some differences tested (§7). Four to six windows per data set give wide intervals across windows, so "level" often means "not separated". Single-fit runs before 16 October trained with seed 0 whatever seed was named.

Studies added after the main results. Studies A to F were added after the rolling results were read. Each was written down before it ran, but which studies to run was chosen knowing that the ensemble came first. Timestral was run at 28M parameters, not 209M, because the larger model was too slow, a change made before any Timestral result.

Comparison models. TFT was not run on M5 or Favorita, and Chronos-Bolt and TimesFM from history alone not on Favorita. TimesFM with the plan fits its linear plan model within groups of 2,048 series. Pretrained models are used as published with their point forecast fixed in advance (the mean), which for Timestral let a few extreme upper quantiles dominate on weekly data. The Timestral tail check and the cheese window were looked at after the results were read.

What lift means. Merchants promote in weeks they expect to sell well, so part of a measured lift is good timing. The lift is what the forecast attributes to the plan, not what the plan caused. Our companion audit [63] measures the gap in generated store data where the true lifts are known: the world model's and LightGBM's lifts were a third too large while every check against the model passed. Plan responses in this paper, including the perturbation tests, are forecasts, not causal estimates.

What the data do not record. None of the data sets records stock, so a sell-out looks like low demand; or shelf space. Favorita records no prices. The data were collected between 1989 and 2017 and their plan inputs differ, so the results compare models within each data set. The decision costs are simulations under stated costs.

10 Conclusion

Which forecasting model works best depends on the grain of the decision it serves. On four public data sets with price and promotion plans, an equal-weight average of a world model, LightGBM and TimesFM given the

plan had the lowest error for items by store and items across stores on most data sets, with few models close to it; for categories and the chain total the best model changed from data set to data set and several could not be told apart; and over a whole window the world model alone gave the lowest category-planning cost on most data sets. LightGBM was the strongest single model at the fine grains, and averaging added to it. These observations cover two of the four Ps on public data; a forecast solved across all four, or a retailer's own data, may order the models differently. The grain comparison itself, run over several windows with paired tests and the losing models stated, is what we propose a retailer repeats on its own data before choosing a model.

11 Reproducibility and compute

Records. All code, configuration files and tests are in the repository. A run refuses to start on uncommitted code and writes a manifest holding the code commit, the full configuration and its checksum, the seed, checksums of every data file read, package versions, the machine and run times. The manifest and scores of the 110 final-window runs are committed under `results/promoted/`. The revision's analyses of saved forecasts are committed as tables under `results/paper1_revision/tables/`, with the per-window scores of study F under `results/paper1_revision/scores/`; the plan and every amendment are in `docs/PAPER1_REVISION_PLAN.md`. `make` in `paper/` rebuilds every figure, table and quoted number from these files (26 seconds from a clean copy of the repository on two CPU cores, 17 October 2026), and a quoted revision number that is missing from them stops the build. One table was transcribed from a console run rather than written by a script: Timestral's quantile sums (`studyF_tails.csv`), from `scripts/timestral_tails.py`. Forecast files and model checkpoints are kept outside the repository with their checksums recorded.

Compute. Every run used one Google Colab machine: an NVIDIA Tesla T4 with 14.6 GB of memory, a 2-core Intel Xeon CPU @ 2.00GHz and 12.7 GB of memory, PyTorch 2.11.0+cu130 with CUDA 13.0. Table 18 gives the run time of every timed final and comparison run of the first version: 94 runs and 44.7 machine-hours. The revision's rolling windows, studies and analyses were not timed as a whole; from the working sessions we estimate about 60 further machine-hours on T4 and L4 GPUs, of which study F took about 12. This is an estimate, not a record.

Table 18: Run times of the final and comparison runs, from their manifests. Wall-clock minutes per run (data loading, training, forecasting and scoring) and hours in total, on one Google Colab machine with an NVIDIA Tesla T4 (LightGBM uses its CPU). Dominick's: six categories together; a Dominick's world-model run averages three fits. Two M5 runs resumed after the Colab runtime dropped record only the resumed part (the shortest values in their rows). Not counted: runs saved before run times were recorded, the development study, and TimesFM on Dominick's and Frat.

Data	Model	Runs	Median	Range	Hours
M5	World model	9	101.5	37.3 to 103.2	14.0
M5	Lift model	9	88.7	10.3 to 94.6	12.2
M5	LightGBM	9	78.1	76.5 to 79.9	11.7
M5	Graph model	12	7.2	7.2 to 7.5	1.5
M5	TimesFM	8	3.9	3.8 to 4.1	0.5
M5	Seasonal naive	2	1.3	1.2 to 1.3	0.0
Dominick's	World model	18	9.2	9.0 to 10.6	2.8
Dominick's	LightGBM	18	4.6	3.1 to 11.5	1.7
Frat	World model	3	2.2	2.2 to 2.2	0.1
Frat	Lift model	3	2.0	1.9 to 2.0	0.1
Frat	LightGBM	3	1.5	1.5 to 1.6	0.1
All 94 timed runs					44.7

12 Data, code and licence

Data. M5 is available from the competition's Kaggle page [1]; Dominick's from the Kilts Center for Marketing, University of Chicago Booth School of Business [19]; Breakfast at the Frat from dunnhumby [20]; Corporación Favorita from its Kaggle competition [67]. Each is obtained from its provider under the provider's own terms. No data is redistributed with the code.

Code. The code is at `https://github.com/Kar488/retail-world-models`.

Licence. The code is licensed under the PolyForm Noncommercial License 1.0.0. The manuscript text, figures and tables are licensed under Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0). Commercial use of either needs a separate licence from the author.

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[69] P. R. Hansen, A. Lunde, and J. M. Nason. The model confidence set. *Econometrica*, 79(2):453–497, 2011.

A Constraints the models work within

Table 19 lists the constraints. The world model's settings are in Table 3.

B Supplement: the development study

This section reports what was tried before the final tests, on the development windows only. It shows how the final models were chosen and which ideas from the literature did not help on these data. Every row is a section of the results record, with its run folders.

B.1 Breakfast at the Frat

Benchmarks. On the three windows before the data set's end, seasonal naive scores 0.832, the average of the last 8 weeks 0.756, LightGBM with price only 0.554 and LightGBM with every recorded lever 0.447. The levers are worth a fifth of the error: display, feature and price reduction carry information price does not.

What helped and what did not. Figure 15 shows every configuration screened on Frat. Our item encoder with a direct readout scored 0.424, level with LightGBM (0.421). Building the forecast as a baseline under the nothing-planned plan times a lift (the lift model) brought it to 0.406, better than LightGBM in all five runs and with half the spread across seeds. Letting the 55 items in a store affect each other, by conditional similarity, attention or both, made the forecast worse in every version (0.443 to 0.476) and less steady; limiting it to items in the same category narrowed the gap without closing it. Extra inputs did not help: the plan of the four weeks before (0.411), the same week last year (0.408), the US seasonal calendar (0.425), the count of items on promotion in the store (0.406) and the recorded base price used as the regular price (0.434). The recorded base price changes about twelve times a year per item and equals the shelf price in every week without a promotion, so it is not a slow-moving regular price.

Latent (JEPA-style) losses. Following the JEPA literature [45, 46] and M-JEPA [57], we added a loss that predicts the encoder's own state for the forecast weeks from the state now and the plan, with a slowly updated copy of the encoder as the target [48]. No recipe beat the lift model: alongside the forecast loss 0.410 to 0.411, pretrained on the latent loss and then fine-tuned 0.407 to 0.414. With the encoder frozen after latent-only pretraining the model still reached 0.428, so a state learned without any sales target carries most of the forecast signal. Long pretraining followed by a slowed encoder gave the best display-week error of any configuration (0.813 for the world model, against 0.817 for LightGBM).

The world model. Rolling the state forward week by week under the plan scored 0.408, level with the lift model on the overall score and on every lever. Three ideas from the world-model literature, each at one untuned setting, did not help: keeping the plan's effect apart from the no-plan state as in DWM [53] (0.417), a latent check on each rolled step with a VICReg penalty [49] and near weeks weighted more as in TD-MPC2 [52] (0.427), and all three together (0.424). Training changes did not help either: stopping at the best point on the last 8 training weeks (0.425) took 8 of only 90 training weeks out of the targets, and a term on summed totals in the loss (0.414) had no bias to fix, since total forecasts were within 3% of sales in every run. Readout dropout with weight decay (0.409) was level and steadier, and was kept for the lift model.

Beyond accuracy. Table 20 reports four further tests. *New items*: with one product in ten held out of training, the lift model borrowing the state and level of the ten known items in the same store with the most labels in common scored 0.890 on the new items, against 1.310 for LightGBM fitted on labels, price and plan, and 1.510 for a forecast of zero. On Dominick's analgesics, where items carry no category or maker labels, borrowing failed (2.999 against 0.923): the method needs item attributes. *Combination*: weighting the lift model and LightGBM on a validation window gave the best Frat result of the study (0.388), better than either part on display and circular weeks, because the two make different errors in promoted weeks; on analgesics the combination was level. Combinations were not carried to the final tests, which compare single models. *Plan A against plan B*: for pairs of weeks of the same item and store with different plans, every model picked the week that sold more about 78% of the time; LightGBM was marginally ahead. *Unusual plans*: on lever mixes an item had run in under 2% of its weeks, no model degraded much and LightGBM stayed ahead. The world model's ability to roll a plan forward did not show as an advantage on these tests.

B.2 Dominick's

Analgesics, the category used to set the models. On analgesics, seasonal naive scored 1.274, the average of the last 8 weeks 0.723 and LightGBM 0.715. The lift model scored 0.676 on three seeds, and the world model 0.667 once a totals term was removed from its loss: that term made the total worse, pulling forecasts 9% low (0.819 for the lift model). Training four times longer

Table 19: Every constraint the trained models and the ensemble work within, with its value and where it lives in the code.

Constraint	Value	Where
Log forecast bound (world and lift model)	$[-10, 10]$ around the item's level	`src/rwm/model/state_model.py` readout
Lift when nothing is planned	Exactly zero	Readout; tested in `tests/`
Baseline under a changed plan	Unchanged	Readout; tested in `tests/`
Regular price	Highest price in the last 12 periods	Plan features
Brand-switching share α	One per run, fitted on the last 8 training weeks	Dominick's world model
Forecasts	Not below zero	Every model
Ensemble weights	One third each, fixed 13 October 2026	`scripts/revision_analysis.py`
Interval pooling	Horizons pooled below 50 earlier errors per horizon	`src/rwm/evaluation/compare.py`
Interval ends	Rounded outwards to whole units (not Favorita)	`src/rwm/evaluation/compare.py`
TimesFM covariate fit	Groups of 2,048 series	`src/rwm/prior_work/`
Timestral context and batch	512 periods; batch 256, halved on running out of GPU memory	`src/rwm/prior_work/timestral_zero_shot.py`

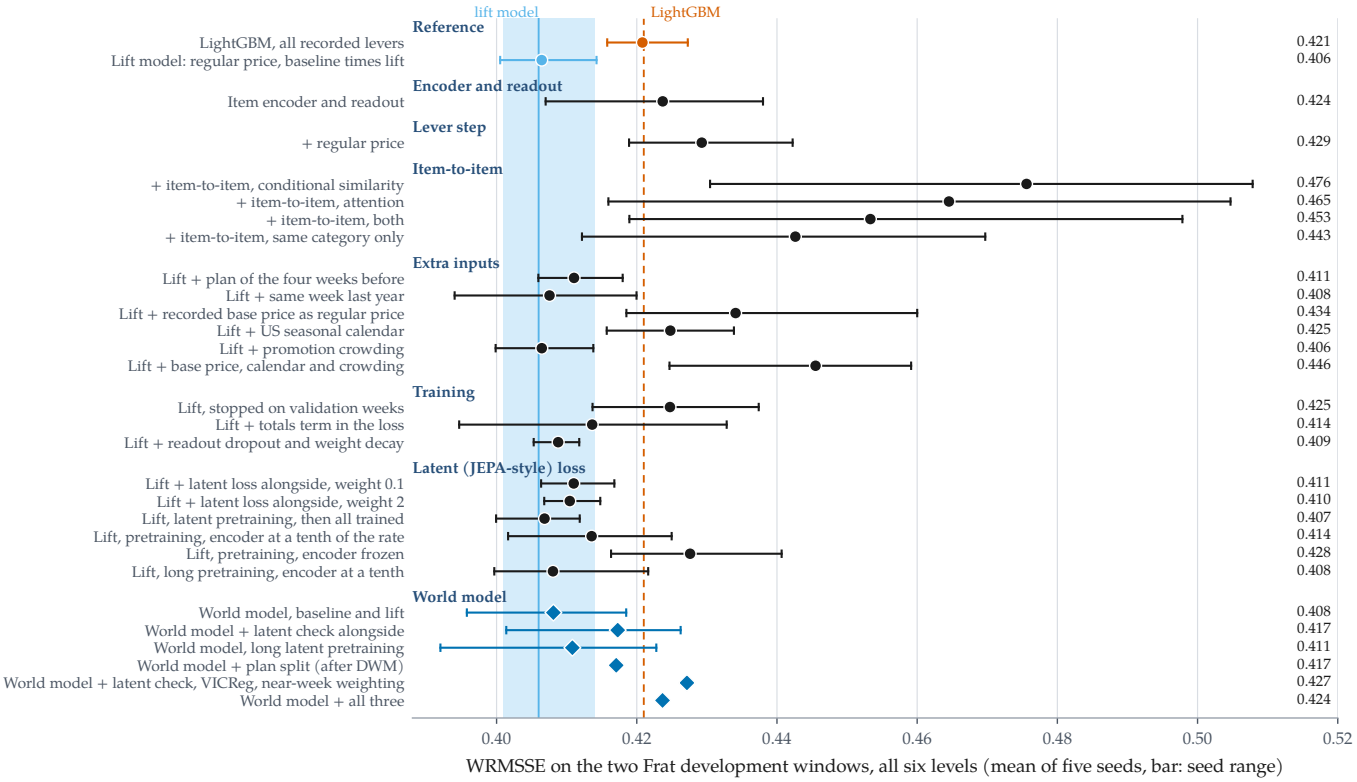


Figure 15: On Frat the gain came from building the forecast as a baseline times a plan lift; item-to-item links, extra inputs, latent losses and the roll-forward did not improve on it. Each row is one configuration on the two development windows, five seeds: mean (marker) and range (bar). Blue band: the lift model's seed range; vertical lines: the lift model (0.406) and LightGBM (0.421). Values below the lift model are in bold. Rows without a bar were recorded as a mean only.

without a stop broke the summed levels (1.473) while item-by-store error barely moved, so a small bias in the same direction on every item added up at the category total. Five seeds changed the picture: the world model held (0.666) and LightGBM scored 0.702, but the lift model fell to 0.718, so a three-seed figure had flattered it.

All 28 categories. Run on all 28 categories with settings from analgesics, LightGBM was the most accurate of the three in 14, the lift model in 5 and the world model in 9, with mean WRMSSE 0.688, 0.784 and 0.747; on promoted weeks LightGBM was most accurate in 24 of 27 (Figure 16, Table 21). Analgesics did not generalise, which is why the paper reports every category. The failures (snack crackers 1.463, bath soap 1.465, the lift

Table 20: What the plan-aware models were also tested on during development: new items, combination with LightGBM, choosing between two plans, and plans an item had rarely run. Development windows, three seeds (Frat combination: five). WRMSSE, lower is better, unless stated. Plan A against plan B: share of week pairs where the forecast picks the week that sold more, and the average miss on the size of the change (log scale).

Data	Model	Result
New items: one product in ten never seen in training (item by store)		
Frat	LightGBM, new item from labels, price and plan	1.310
Frat	Lift model, borrowing from similar items	**0.890**
Frat	Forecast of zero	1.510
Analgesics	LightGBM, from labels, price and plan	**0.923**
Analgesics	Lift model, borrowing (no attribute labels)	2.999
Combination with LightGBM, weights set on a validation window (all levels)		
Frat	Lift model + LightGBM	**0.388**
Frat	World model + LightGBM	0.390
Frat	LightGBM; lift model; world model	0.421; 0.406; 0.408
Analgesics	Lift + LightGBM; world + LightGBM	0.670; 0.668
Plan A against plan B: same item and store, two weeks with different plans (Frat)		
Frat	LightGBM	78.1% right; miss 0.448
Frat	Lift model	77.4% right; miss 0.460
Frat	World model	77.5% right; miss 0.459
Frat	World model + plan split	77.5% right; miss 0.458
Unusual plans: a lever mix the item had in under 2% of its weeks (item by store)		
Frat	LightGBM	0.696 unusual; 0.697 usual
Frat	Lift model	0.742 unusual; 0.725 usual
Frat	World model	0.730 unusual; 0.730 usual

model on soft drinks 1.406) sat at the category total: item-by-store error was close to LightGBM's. On a six-store sample of snack crackers the cause was in-and-out and seasonal lines: items recorded in 8 to 20 of the last 26 weeks were 8% of sales and the lift model forecast them at 9 times their sales, because the model set each item's level from its 26-week average after the item had fallen away. Capping extreme weeks did little; taking the level from the last 8 weeks removed most of it. On all 28 categories the 8-week level brought the world model's mean to 0.717, removed the large failures (snack crackers 1.463 to 0.874, bath soap 1.465 to 1.195) and gave a little back in categories that had been fine, so it beat LightGBM in 10 categories against 14 before. LightGBM, which sees short recent averages directly, does not have the problem.

Peaks and dips. In cheese, both our model and LightGBM under-forecast promoted weeks by about a third in the holiday window, and cheese peaks do not sit on a fixed week relative to the holidays: they follow the chain's promotion timing, which event flags cannot place. Figure 17 splits forecast against sales into promoted weeks, the weeks just after a promotion, and ordinary weeks, for six categories. The world model did not see the dip after a promotion where the dip is large: it forecast 1.33 times sales after a promotion in snack crackers and 1.32 in soft drinks, against 0.92 and 0.90 for LightGBM. Feeding the model's own sales forecast into the roll-forward, so that a promotion's sales could shape the weeks after it, fixed the soft-drink dip (1.32 to 1.00) and made the overall score worse in four of six categories. Weighting the loss by dollar sales and by how far a week sits from the item's usual level fixed cheese promoted weeks (1.21 to 1.00) and was worse overall in three categories. In frozen entrees every model, LightGBM included, forecast about 0.7 of sales after a promotion: there the week after is stronger than any model expected.

From the first world model to the one tested. Figure 18 follows the world model on the six categories through the changes kept, three seeds each. Across seeds the first version was unsteady: its range within a category was 0.04 to 0.38, against under 0.03 for LightGBM, and on the mean LightGBM was ahead in five of six. Brand switching fixed what it was built for: in snack crackers and soft drinks, where promoted weeks broke the total, it was better on every seed (0.813 to 0.752, 0.902 to 0.789) and item-by-store error improved too. The fitted share of the summed lift that the category total keeps (Figure 18g) is 0.00 to 0.66 in snack crackers, against about 0.9 in bot-

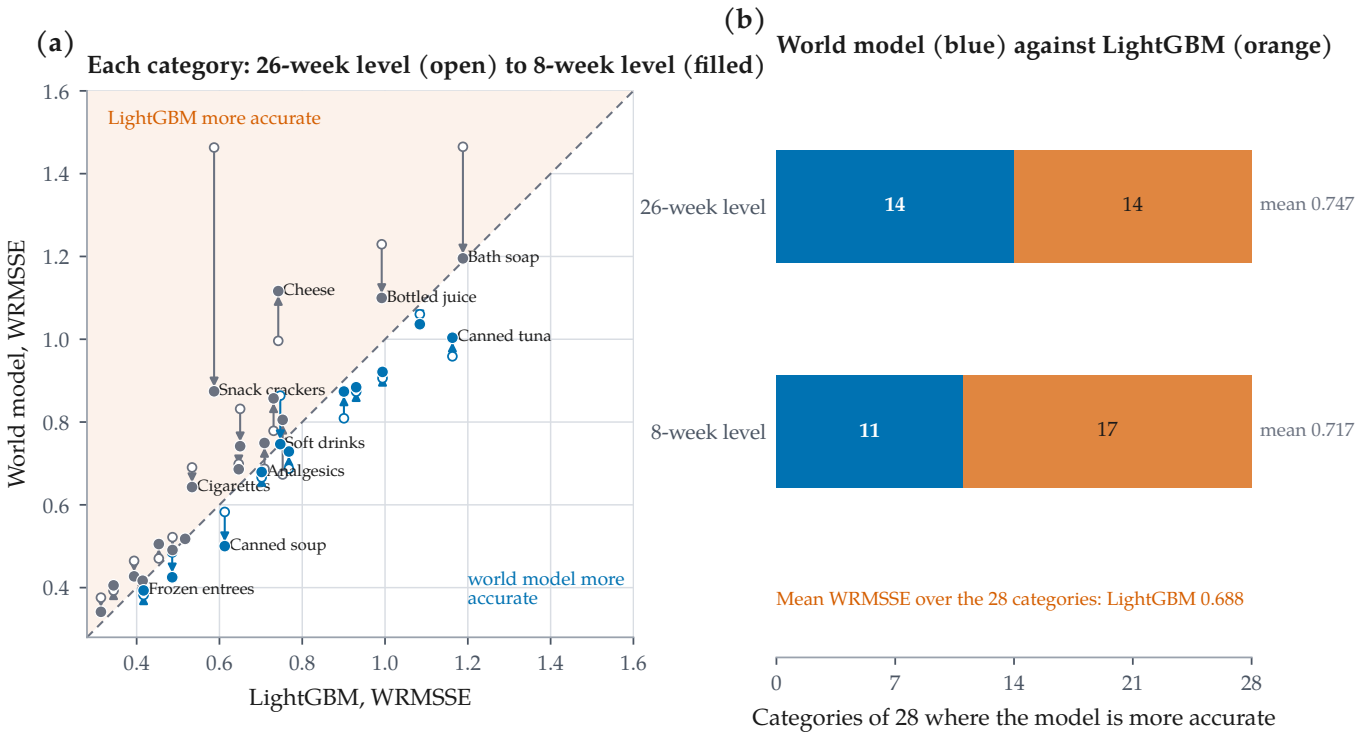


Figure 16: Run unchanged on all 28 Dominick's categories, the world model set on analgesics failed badly in a few; taking each item's level from its last 8 weeks removed the worst failures but did not make it better than LightGBM on average. Development windows, one seed. **(a)** Each category's WRMSSE for LightGBM (horizontal) and the world model (vertical), with the item level from the last 26 weeks (open) and the last 8 weeks (filled); blue: the world model is more accurate with the 8-week level. **(b)** Categories where each model is more accurate, and the mean over the 28.

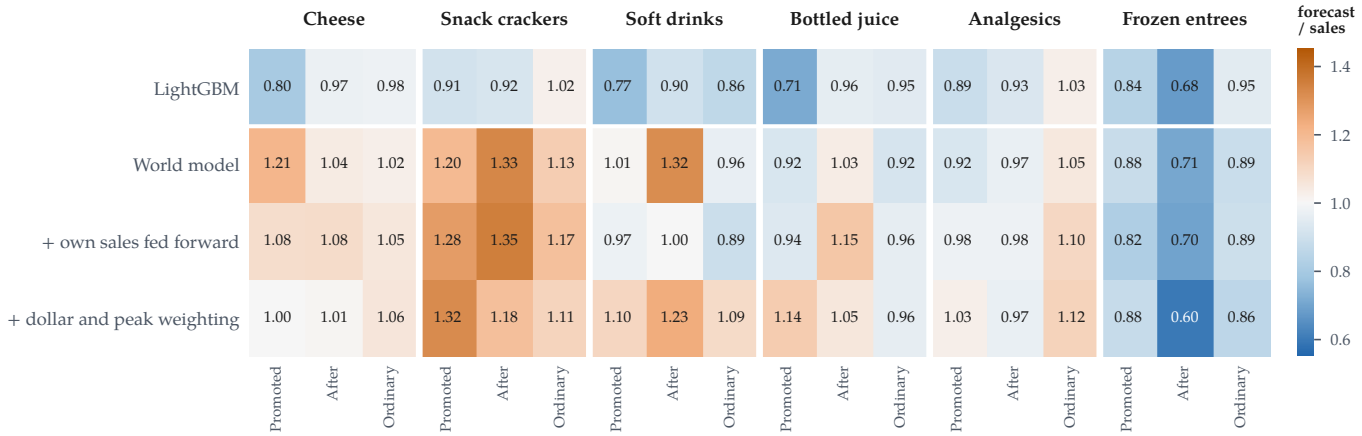


Figure 17: The world model over-forecast the weeks after a promotion where shoppers had stocked up; the two fixes each repaired one category and hurt others. Forecast against sales (1.0 is right; orange over, blue under) in promoted weeks, the weeks just after a promotion and ordinary weeks, Dominick's development windows, one seed. Rows: LightGBM, the world model with the 8-week level, with its own sales forecast fed forward, and with the loss weighted by dollar sales and by peaks.

tled juice, analgesics and frozen entrees: in snack crackers the summed item lifts overstated the category total, and the fit took most of it back. Where it comes out at 1.0 the step changes nothing, a built-in check. The direction agrees with van Heerde, Gupta and Wittink's household estimates [15], but the share is fitted to make totals right; it is not an estimate of switching. Normalising the rolled state narrowed the seed range in five of six categories, and averaging three fits per run with normalisation gave the steadiest version (range 0.05 or less in five categories) and the candidate for the final test: better than LightGBM on the three-seed mean in analgesics (0.621 against 0.701), frozen entrees (0.391 against 0.419) and soft drinks (0.743 against 0.759). One snack-cracker run of the three failed (1.806): its second window's category total broke with promoted weeks forecast at 2.6

Table 21: Across all 28 Dominick's categories LightGBM is more accurate on average; the world model wins in 14 with its first level and 10 with the 8-week level. Development windows, one seed (analgesics: five), settings set on analgesics and not tuned per category. World model with the item's level from its last 26 weeks or last 8 weeks. Lowest in each row in bold.

Category	LightGBM	Lift	World, 26 wk	World, 8 wk
Analgesics	0.702	0.718	**0.666**	0.679
Bath soap	**1.188**	1.568	1.465	1.195
Beer	**0.454**	0.460	0.470	0.505
Bottled juice	**0.992**	1.247	1.229	1.100
Canned soup	0.613	0.532	0.583	**0.500**
Canned tuna	1.163	1.030	**0.959**	1.004
Cereals	0.753	0.795	**0.673**	0.805
Cheese	**0.742**	0.900	0.996	1.116
Cigarettes	**0.534**	0.918	0.690	0.643
Cookies	**0.487**	0.503	0.522	0.491
Crackers	0.709	0.967	**0.687**	0.749
Dish detergent	**0.344**	0.436	0.393	0.405
Fabric softener	0.414	**0.358**	0.397	0.417
Front-end candies	**0.650**	0.748	0.832	0.742
Frozen dinners	0.930	0.905	**0.873**	0.884
Frozen entrees	0.417	**0.374**	0.382	0.393
Frozen juices	**0.314**	0.360	0.376	0.342
Grooming products	0.901	1.208	**0.809**	0.874
Laundry detergent	0.768	0.735	**0.687**	0.729
Oatmeal	1.084	**1.026**	1.060	1.036
Paper towels	0.486	0.445	0.485	**0.425**
Shampoos	0.994	0.938	**0.906**	0.921
Snack crackers	**0.587**	0.959	1.463	0.874
Soaps	0.517	0.521	**0.516**	0.518
Soft drinks	0.747	1.406	0.864	**0.747**
Toilet tissue	**0.394**	0.413	0.465	0.427
Toothbrushes	**0.647**	0.671	0.699	0.686
Toothpaste	**0.731**	0.802	0.779	0.857
Mean	0.688	0.784	0.747	0.717

times sales while item-by-store error stayed normal, so the summed promotion lift can still run away on one fit in three. Twice the training steps, screened after the final tests were fixed, was better in two categories, worse in two, level in one and mixed in one, and was not kept.

Other changes. Showing the model both its 26-week and its recent levels as inputs, instead of setting its scale by the recent level, was the best world-model version in four of ten categories but did not fix snack crackers (1.515) or soft drinks (1.069): the model did not learn to switch to the recent level when the long level still set its scale. Item role labels (new line, in-and-out, volume driver, long tail, high-low, promotion-responsive, core), assigned from each item's own sales before the windows, moved LightGBM by at most 0.013, since it reads the same information from its history, and moved the world model in both directions; letting the role pick the level broke snack crackers (2.422). We did not call the volume drivers key value items, since a key-value-item list rests on shopper price perception, which these data do not record.

Table 22: On the M5 practice window, only the normalised roll-forward separated the world model from LightGBM; no later change beat it. WRMSSE over all 12 levels, 25 April to 22 May 2016. A screened change was kept only if it lowered the score, and a change that doubles the cost only if by more than the seed range (0.029).

Model or change	Mean	Seed range
Seasonal naive	0.870	
LightGBM, one model per store	0.715	
Lift model (3 seeds)	0.713	0.683 to 0.734
Item encoder and readout (5 seeds)	0.699	0.669 to 0.724
World model (3 seeds)	0.707	0.679 to 0.750
World model + normalised roll-forward (3 seeds)	0.656	0.646 to 0.675
+ average of three fits (1 run)	0.665	
+ weekday as seven inputs (seed 1)	0.653	
+ weekday and event kinds (seed 1)	0.646	
+ 60,000 training steps (seed 1)	0.642	
+ item level against past-year level (seed 1)	0.680	

B.3 M5

On the M5 practice window (Table 22), the item encoder with a direct readout scored 0.699 over five seeds against 0.715 for LightGBM, and the lift and world models were level with LightGBM over three seeds (0.713 and 0.707). Normalising the rolled state was the one change that separated the world model: 0.656, with its worst seed (0.675) better than the best seed without it (0.679), all of the gain at the totals and the seed range cut from 0.071 to 0.029. It helped on M5's 28 daily steps and not on Dominick's 8 weekly steps, which fits the reading that it stops the state drifting over a long roll. Averaging three fits did not help on M5 (0.665), where single fits were already steady.

After the official test month, an error analysis of the world model found its 2.6% excess in Foods (+3.9%), in one store (CA_3, +7.5%), and on Sundays and Mondays (+5.1%, +6.4%), not growing with the horizon. Grouped by sales in the 28 days before the test, the excess came from items that had been selling fast (+5.4% and +6.1% on items selling 3 to 10 and over 10 a day). Four changes were screened on the practice window under rules written before screening: weekday as seven inputs, event kinds, twice the training steps, and the item's level against its past-year level. None beat the original (Table 22), so there was no second version. On the practice window every version under-forecast by 4% to 6%, while on the test month the world model over-forecast

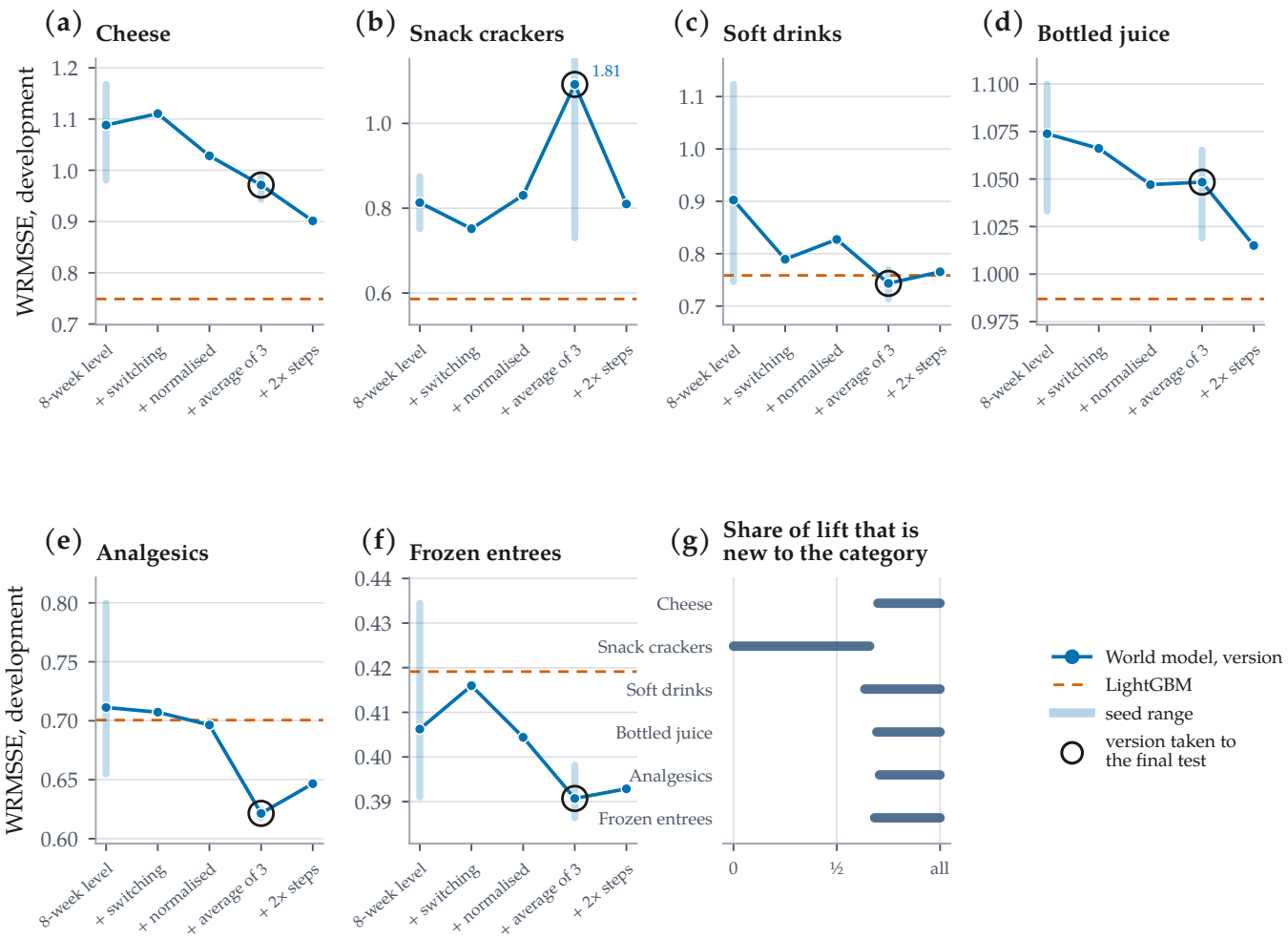


Figure 18: Brand switching, a normalised roll-forward and averaging three fits took the world model past LightGBM in three of six categories on the development windows, and the switching step learned how much of a promotion's lift is new to the category. (**a–f**) Development WRMSSE of each version, three-seed means; shaded bars: seed range of the first version and of the one taken to the final test (circled); dashed: LightGBM. (**g**) The fitted share of the summed promotion lift that the category total keeps, the split that makes category totals fit best (not an estimate of switching), range over runs.

by 2.6%: the sign of the level error changed between adjacent months, so a fix tuned on one would not carry to the next. That finding is what led to the confirmation months of §6.10.